

CS 690U Project 2025

Diverse Genome Embedding Benchmark

Declaration of Intent: \

Basic Idea: Compare a simple model with a complex model from the DGEB paper in case of EC-Classification -> do we really need a complex foundation model?

Using AA (amino acids)

I will test or plot: \

The performance (e.g., F1-score) of a logistic regression model trained on one-hot encoded protein sequences for EC number prediction. I will also compare this performance to published results from foundation models in the DGEB benchmark.

If the hypothesis is true, I will observe: \

The logistic regression model will achieve comparable performance (in terms of F1-score) to the foundation model baseline on the EC Classification task, indicating that simple models can be effective with basic encodings.

If the hypothesis is false, I will observe:

The logistic regression model will perform significantly worse than the foundation models, suggesting that the complex, pre-trained representations are capturing important functional features that simple models miss.

Idea:

Getting TSV -> Do One-Hot-Encoding -> Split -> train Modell -> test Model -> compare Results with DGEB -> discussion

```
In [ ]: #! pip install biopython scikit-learn pandas numpy
        #! pip install dgeb
        #! pip uninstall -y scipy
        #! pip uninstall -y scikit-learn

        # ! pip install scipy scikit-learn --force-reinstall --no-cache-dir
```

```

Defaulting to user installation because normal site-packages is not
writeable
Collecting scipy
  Downloading scipy-1.13.1-cp39-cp39-macosx_12_0_arm64.whl.metadata
(60 kB)
Collecting scikit-learn
  Downloading scikit_learn-1.6.1-cp39-cp39-macosx_12_0_arm64.whl.met
adata (31 kB)
Collecting numpy<2.3,>=1.22.4 (from scipy)
  Downloading numpy-2.0.2-cp39-cp39-macosx_14_0_arm64.whl.metadata (
60 kB)
Collecting joblib>=1.2.0 (from scikit-learn)
  Downloading joblib-1.4.2-py3-none-any.whl.metadata (5.4 kB)
Collecting threadpoolctl>=3.1.0 (from scikit-learn)
  Downloading threadpoolctl-3.6.0-py3-none-any.whl.metadata (13 kB)
Downloading scipy-1.13.1-cp39-cp39-macosx_12_0_arm64.whl (30.3 MB)
_____ 30.3/30.3 MB 5.4 MB/s et
a 0:00:0000:0100:01
Downloading scikit_learn-1.6.1-cp39-cp39-macosx_12_0_arm64.whl (11.1
MB)
_____ 11.1/11.1 MB 9.9 MB/s et
a 0:00:00a 0:00:01
Downloading joblib-1.4.2-py3-none-any.whl (301 kB)
Downloading numpy-2.0.2-cp39-cp39-macosx_14_0_arm64.whl (5.3 MB)
_____ 5.3/5.3 MB 11.4 MB/s eta
0:00:00a 0:00:01
Downloading threadpoolctl-3.6.0-py3-none-any.whl (18 kB)
Installing collected packages: threadpoolctl, numpy, joblib, scipy,
scikit-learn
  Attempting uninstall: threadpoolctl
    Found existing installation: threadpoolctl 3.6.0
    Uninstalling threadpoolctl-3.6.0:
      Successfully uninstalled threadpoolctl-3.6.0
  Attempting uninstall: numpy
    Found existing installation: numpy 2.0.2
    Uninstalling numpy-2.0.2:
      Successfully uninstalled numpy-2.0.2
  Attempting uninstall: joblib
    Found existing installation: joblib 1.4.2
    Uninstalling joblib-1.4.2:
      Successfully uninstalled joblib-1.4.2
  Attempting uninstall: scipy
    Found existing installation: scipy 1.13.1
    Uninstalling scipy-1.13.1:
      Successfully uninstalled scipy-1.13.1
  Attempting uninstall: scikit-learn
    Found existing installation: scikit-learn 1.6.1
    Uninstalling scikit-learn-1.6.1:
      Successfully uninstalled scikit-learn-1.6.1
^C
ERROR: Operation cancelled by user

```

```

In [ ]: import scipy
        print(scipy.__version__)
        from sklearn.metrics import classification_report, ConfusionMatrixD
        from transformers import AutoTokenizer, AutoModel

```

```
import torch
import pandas as pd
import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report
import matplotlib.pyplot as plt
from sklearn.metrics import ConfusionMatrixDisplay
```

1.13.1

Preperation of the Dataset

Load Dataset

UniProt Dataset

```
In [ ]: tsv_file = "uniprotkb_ec_AND_reviewed_true_AND_frag_2025_04_10.tsv"
df = pd.read_csv(tsv_file, sep='\t')

# Just keep the relevant columns with exact one EC number
df = df[df['EC number'].notna()]
df = df[df['EC number'].apply(lambda x: ";" not in str(x))]
# print(df)

# just keep classes with at least 5 sequences
df['EC_class'] = df['EC number']
class_counts = df['EC_class'].value_counts()
print(f"Anzahl:", class_counts)
valid_classes = class_counts[class_counts >= 5].index
df = df[df['EC_class'].isin(valid_classes)]

# Fit sequence length
max_len = 256 #512
aa_list = list("ACDEFGHIKLMNPQRSTVWY")
def clean_sequence(seq):
    seq = seq[:max_len]
    seq = ''.join([aa if aa in aa_list else 'X' for aa in seq])
    return seq.ljust(max_len, 'X')
df["CleanedSeq"] = df["Sequence"].apply(clean_sequence)

print(df["CleanedSeq"].head())
```

```

Anzahl: EC_class
7.1.1.-          4626
2.7.7.6          3943
3.1.-.-          3902
2.7.11.1         2420
2.1.1.-          2326
...
1.13.11.9        1
1.14.14.158      1
6.2.1.46         1
4.2.3.167        1
4.1.1.76         1
Name: count, Length: 5176, dtype: int64
1    MRFQVIVAAATITMITSYIPGVASQSTSDGDDLFPVPSNFDPKSIF...
3    MYRKLAVISAFLATARAQSACTLQSEHPPLTWQKSSGGTCTQQT...
4    MRSAILTTLLAGHAFAYPKPAPQSVNRRDWPSINEFLSELAKVMP...
5    MIVGILTTLATLATLAASVPLEERQACSSVWGQCGGQNWSGPTCCA...
6    MYWQLVRILVLFDCLOKILAIEHDSICIADVDDACPEPSHTVMRLR...
Name: CleanedSeq, dtype: object

```

```

In [3]: # Note: 2378 classes were too much to train the model ( time > 1h)
top_classes = df['EC_class'].value_counts().head(100).index
df = df[df['EC_class'].isin(top_classes)]

```

Hot-Encoding

```

In [4]: # One-Hot-Encoding
def one_hot_encode_sequence(seq):
    matrix = np.zeros((max_len, 21))
    for i, aa in enumerate(seq):
        idx = aa_list.index(aa) if aa in aa_list else 20
        matrix[i, idx] = 1
    return matrix.flatten()

X = np.stack(df["CleanedSeq"].apply(one_hot_encode_sequence))
y = df["EC_class"].values

print("Dataset successfully computed.")
print("Input shape:", X.shape)
print("Number of classes:", len(set(y)))

```

```

Dataset successfully computed.
Input shape: (95590, 5376)
Number of classes: 100

```

Split-up Dataset

```

In [5]: # Split the data into training and test sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
X_train_customDS = X_train
X_test_customDS = X_test
y_train_customDS = y_train

```

```
y_test_customDS = y_test
```

Logistic Regression

Setup and train model

```
In [ ]: model = LogisticRegression(max_iter=500, solver='saga', multi_class
# Klassifikationsmodell trainieren
model.fit(X_train, y_train)
```

/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/sklearn/linear_model/_logistic.py:1247: FutureWarning: 'multi_class' was deprecated in version 1.5 and will be removed in 1.7. From then on, it will always use 'multinomial'. Leave it to its default value to avoid this warning.

```
warnings.warn(
Epoch 1, change: 1
Epoch 2, change: 0.3000886
Epoch 3, change: 0.17558489
```

```
In [ ]: #better safe than sorry
np.save('X_train_UniProt.npy', X_train)
np.save('X_test_UniProt.npy', X_test)
np.save('y_train_UniProt.npy', y_train)
np.save('y_test_UniProt.npy', y_test)
```

Result evaluation

```
In [ ]: y_pred = model.predict(X_test)
print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
1.14.14.18	0.00	0.00	0.00	1
1.15.1.1	0.00	0.00	0.00	1
1.16.3.1	0.00	0.00	0.00	1
1.17.4.1	0.00	0.00	0.00	1
1.5.98.3	0.00	0.00	0.00	1
1.8.3.2	0.00	0.00	0.00	1
2.1.1.354	0.00	0.00	0.00	1
2.1.1.360	0.00	0.00	0.00	1
2.1.1.37	1.00	1.00	1.00	1
2.1.1.72	0.00	0.00	0.00	1
2.1.1.77	0.33	1.00	0.50	1
2.1.1.86	1.00	1.00	1.00	1
2.1.3.15	0.00	0.00	0.00	1
2.3.1.225	0.00	0.00	0.00	1
2.3.1.269	0.33	1.00	0.50	1
2.3.1.48	0.00	0.00	0.00	1
2.3.1.51	0.50	1.00	0.67	1
2.3.2.23	1.00	1.00	1.00	1
2.3.2.26	0.00	0.00	0.00	1
2.3.2.27	0.00	0.00	0.00	1

2.3.2.31	0.00	0.00	0.00	1
2.4.1.109	0.00	0.00	0.00	1
2.4.1.198	0.00	0.00	0.00	1
2.4.2.31	0.00	0.00	0.00	1
2.5.1.18	0.00	0.00	0.00	1
2.7.1.107	0.00	0.00	0.00	1
2.7.1.21	0.00	0.00	0.00	1
2.7.1.33	0.50	1.00	0.67	1
2.7.1.67	0.00	0.00	0.00	1
2.7.10.1	0.00	0.00	0.00	1
2.7.10.2	0.00	0.00	0.00	1
2.7.11.1	0.00	0.00	0.00	1
2.7.11.25	0.00	0.00	0.00	1
2.7.13.3	0.00	0.00	0.00	1
2.7.4.3	0.25	1.00	0.40	1
2.7.6.1	1.00	1.00	1.00	1
2.7.7.108	1.00	1.00	1.00	1
2.7.7.19	0.00	0.00	0.00	1
2.7.7.48	0.00	0.00	0.00	1
2.7.7.49	0.00	0.00	0.00	1
2.7.7.6	0.00	0.00	0.00	1
2.7.7.65	0.00	0.00	0.00	1
2.7.7.7	0.50	1.00	0.67	1
2.7.8.7	1.00	1.00	1.00	1
2.8.1.13	1.00	1.00	1.00	1
3.1.1.1	0.00	0.00	0.00	1
3.1.1.3	0.00	0.00	0.00	1
3.1.1.32	0.00	0.00	0.00	1
3.1.1.4	0.00	0.00	0.00	1
3.1.1.96	0.00	0.00	0.00	1
3.1.11.2	0.00	0.00	0.00	1
3.1.11.6	0.00	0.00	0.00	1
3.1.12.1	0.33	1.00	0.50	1
3.1.13.4	0.00	0.00	0.00	1
3.1.21.10	0.00	0.00	0.00	1
3.1.21.4	0.00	0.00	0.00	1
3.1.26.3	0.33	1.00	0.50	1
3.1.26.4	0.00	0.00	0.00	1
3.1.26.5	0.00	0.00	0.00	1
3.1.3.16	0.00	0.00	0.00	1
3.1.3.18	0.50	1.00	0.67	1
3.1.3.2	1.00	1.00	1.00	1
3.1.3.3	0.00	0.00	0.00	1
3.1.3.48	0.00	0.00	0.00	1
3.1.3.5	0.00	0.00	0.00	1
3.1.4.12	0.00	0.00	0.00	1
3.1.4.35	0.00	0.00	0.00	1
3.1.4.4	0.00	0.00	0.00	1
3.1.4.52	0.00	0.00	0.00	1
3.2.1.1	0.00	0.00	0.00	1
3.2.1.14	0.00	0.00	0.00	1
3.2.1.17	0.00	0.00	0.00	1
3.2.1.18	0.00	0.00	0.00	1
3.2.1.22	0.00	0.00	0.00	1
3.2.1.23	0.00	0.00	0.00	1
3.2.1.35	0.00	0.00	0.00	1

3.2.1.39	0.00	0.00	0.00	1
3.2.1.4	0.00	0.00	0.00	1
3.2.1.52	0.00	0.00	0.00	1
3.2.1.55	0.00	0.00	0.00	1
3.2.1.78	0.00	0.00	0.00	1
3.2.1.8	1.00	1.00	1.00	1
3.2.1.96	0.00	0.00	0.00	1
3.2.2.22	0.00	0.00	0.00	1
3.2.2.6	0.00	0.00	0.00	1
3.4.19.12	0.00	0.00	0.00	1
3.4.21.105	0.00	0.00	0.00	1
3.4.22.49	0.00	0.00	0.00	1
3.4.23.36	0.25	1.00	0.40	1
3.5.1.28	0.00	0.00	0.00	1
3.5.1.98	0.00	0.00	0.00	1
3.5.2.6	0.00	0.00	0.00	1
3.6.1.1	0.00	0.00	0.00	1
3.6.1.13	0.33	1.00	0.50	1
3.6.1.23	0.00	0.00	0.00	1
3.6.1.5	0.00	0.00	0.00	1
3.6.1.9	0.00	0.00	0.00	1
3.6.4.12	0.00	0.00	0.00	1
3.6.4.13	0.00	0.00	0.00	1
3.8.1.2	1.00	1.00	1.00	1
4.2.1.1	0.00	0.00	0.00	1
4.2.1.10	0.00	0.00	0.00	1
4.2.1.2	0.00	0.00	0.00	1
4.2.1.20	0.00	0.00	0.00	1
4.2.2.2	0.00	0.00	0.00	1
4.2.2.3	0.00	0.00	0.00	1
4.2.2.n1	1.00	1.00	1.00	1
4.4.1.3	0.00	0.00	0.00	1
4.6.1.1	0.00	0.00	0.00	1
4.6.1.16	0.00	0.00	0.00	1
4.6.1.2	0.00	0.00	0.00	1
5.2.1.8	0.00	0.00	0.00	1
5.3.4.1	0.00	0.00	0.00	1
5.4.99.5	0.00	0.00	0.00	1
5.6.2.2	0.00	0.00	0.00	1
5.6.2.4	0.00	0.00	0.00	1
6.1.1.19	0.00	0.00	0.00	1
6.3.2.4	0.25	1.00	0.40	1
6.3.4.19	0.00	0.00	0.00	1
6.3.5.2	1.00	1.00	1.00	1
6.3.5.5	1.00	1.00	1.00	1
6.5.1.1	0.00	0.00	0.00	1
6.5.1.3	0.00	0.00	0.00	1
7.1.1.2	0.00	0.00	0.00	1
7.1.1.8	0.00	0.00	0.00	1
7.1.1.9	0.00	0.00	0.00	1
7.1.2.2	0.00	0.00	0.00	1
7.2.1.1	0.00	0.00	0.00	1
accuracy			0.20	128
macro avg	0.14	0.20	0.15	128
weighted avg	0.14	0.20	0.15	128

```

/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/s
klearn/metrics/_classification.py:1565: UndefinedMetricWarning: Prec
ision is ill-defined and being set to 0.0 in labels with no predi
cted samples. Use `zero_division` parameter to control this behavior.
    _warn_prf(average, modifier, f"{metric.capitalize()} is", len(resu
lt))
/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/s
klearn/metrics/_classification.py:1565: UndefinedMetricWarning: Prec
ision is ill-defined and being set to 0.0 in labels with no predi
cted samples. Use `zero_division` parameter to control this behavior.
    _warn_prf(average, modifier, f"{metric.capitalize()} is", len(resu
lt))
/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/s
klearn/metrics/_classification.py:1565: UndefinedMetricWarning: Prec
ision is ill-defined and being set to 0.0 in labels with no predi
cted samples. Use `zero_division` parameter to control this behavior.
    _warn_prf(average, modifier, f"{metric.capitalize()} is", len(resu
lt))

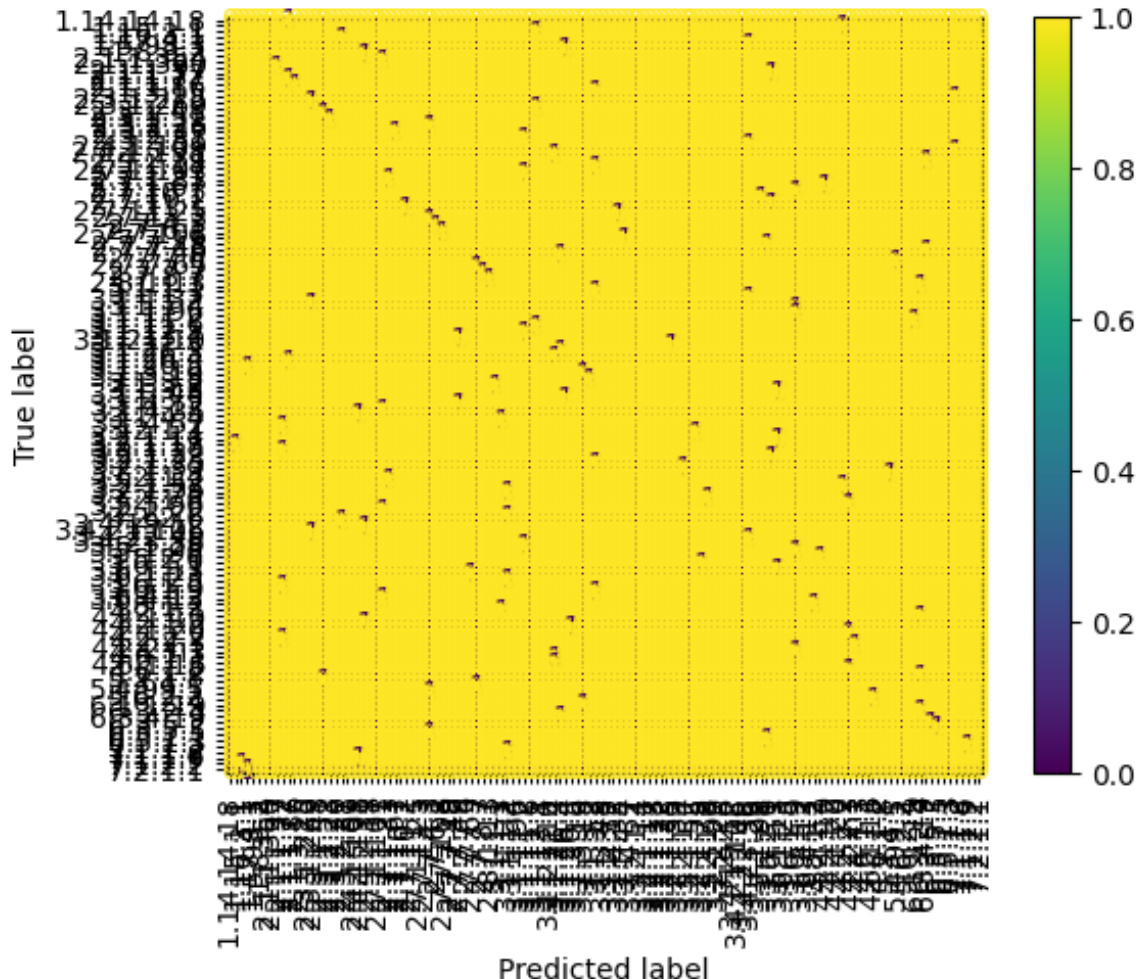
```

Test Model performance

```

In [ ]: # Confusion Matrix
ConfusionMatrixDisplay.from_estimator(model, X_test, y_test)
plt.xticks(rotation=90)
plt.show()

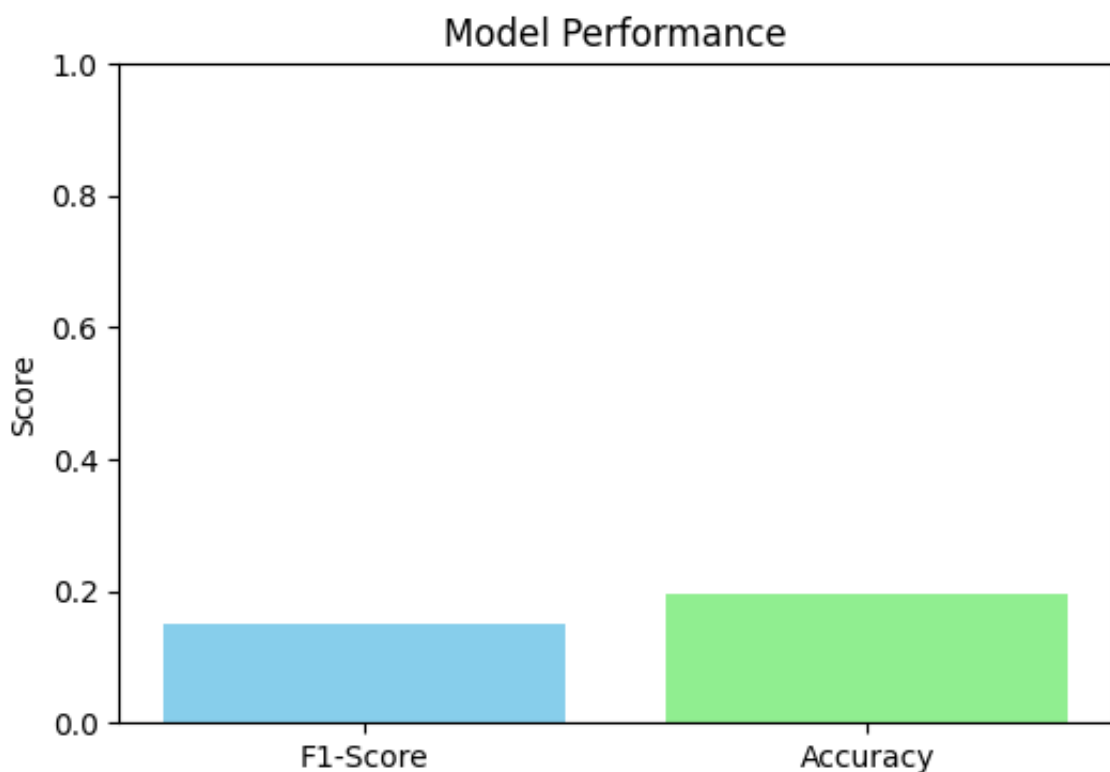
```



```
In [ ]: #Ergebnis-Plot: F1-Score und Accuracy
f1 = f1_score(y_test, y_pred, average='weighted')
acc = accuracy_score(y_test, y_pred)

plt.figure(figsize=(6,4))
plt.bar(['F1-Score', 'Accuracy'], [f1, acc], color=['skyblue', 'lightgreen'])
plt.ylim(0,1)
plt.title('Model Performance')
plt.ylabel('Score')
plt.show()

print("F1-Score:", f1)
print("Accuracy:", acc)
```



F1-Score: 0.15130208333333334

Accuracy: 0.1953125

k-mer method increases the result from F1-score of 0.057 and accuracy of 8.5% to F1-score 0.1513 and accuracy = 19.53%

```
In [ ]: # Save model
import joblib
joblib.dump(model, 'ec_model.joblib')
print("Model saved as ec_model.joblib")
```

Model saved as ec_model.joblib

DGEB Dataset

Load Dataset from Huggingface

```
In [15]: from datasets import load_dataset

dataset = load_dataset("tattabio/ec_classification")
train_data = dataset["train"]
test_data = dataset["test"]
```

Hot-Endcoding

```
In [ ]: # define acid_list
aa_list = list("ACDEFGHIKLMNPQRSTVWY")
max_len = 256 #512

def clean_and_encode(seq):
    seq = seq[:max_len]
    seq = ''.join([aa if aa in aa_list else 'X' for aa in seq])
    seq = seq.ljust(max_len, 'X')
    one_hot = np.zeros((max_len, 21))
    for i, aa in enumerate(seq):
        idx = aa_list.index(aa) if aa in aa_list else 20
        one_hot[i, idx] = 1
    return one_hot.flatten()
```

Prepare Dataset

```
In [ ]: # trainng dataset
X_train = np.stack([clean_and_encode(x['Sequence']) for x in train_
y_train = [x['Label'] for x in train_data]

# test dataset
X_test = np.stack([clean_and_encode(x['Sequence']) for x in test_da
y_test = [x['Label'] for x in test_data]

print("Trainingsdaten:", X_train.shape, y_train[:5])
print("Testdaten:", X_test.shape, y_test[:5])

print(len(X_train))
print(len(X_test))
print(len(y_train))
print(len(y_test))
```

```
Trainingsdaten: (512, 5376) ['1.14.14.18', '1.14.14.18', '1.14.14.1
8', '1.14.14.18', '1.15.1.1']
Testdaten: (128, 5376) ['1.14.14.18', '1.15.1.1', '1.16.3.1', '1.17.
4.1', '1.5.98.3']
512
128
512
128
```

Logistic Regression

Setup and train model

```
In [19]: from sklearn.linear_model import LogisticRegression

model = LogisticRegression(max_iter=500, solver='saga', multi_class
# Klassifikationsmodell trainieren
model.fit(X_train, y_train)
```

/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/sklearn/linear_model/_logistic.py:1247: FutureWarning: 'multi_class' was deprecated in version 1.5 and will be removed in 1.7. From then on, it will always use 'multinomial'. Leave it to its default value to avoid this warning.

```
warnings.warn(
Epoch 1, change: 1
Epoch 2, change: 0.43071083
Epoch 3, change: 0.30358071
Epoch 4, change: 0.27010064
Epoch 5, change: 0.19719495
Epoch 6, change: 0.14770431
Epoch 7, change: 0.092806506
Epoch 8, change: 0.077321722
Epoch 9, change: 0.0455733
Epoch 10, change: 0.10647889
Epoch 11, change: 0.062587379
Epoch 12, change: 0.062503919
Epoch 13, change: 0.037817683
Epoch 14, change: 0.030875078
Epoch 15, change: 0.026905791
Epoch 16, change: 0.025366777
Epoch 17, change: 0.024550922
Epoch 18, change: 0.013840921
Epoch 19, change: 0.011564019
Epoch 20, change: 0.0093062622
Epoch 21, change: 0.0096978461
Epoch 22, change: 0.0084029306
Epoch 23, change: 0.0081908132
Epoch 24, change: 0.0072542217
Epoch 25, change: 0.0068421576
Epoch 26, change: 0.0068877101
Epoch 27, change: 0.006885632
Epoch 28, change: 0.0056290849
Epoch 29, change: 0.0059117078
Epoch 30, change: 0.0050250547
Epoch 31, change: 0.0046516523
Epoch 32, change: 0.0041534537
Epoch 33, change: 0.0041443156
Epoch 34, change: 0.0039978837
Epoch 35, change: 0.0033257103
Epoch 36, change: 0.0034088284
Epoch 37, change: 0.0032266301
Epoch 38, change: 0.0031914127
Epoch 39, change: 0.0031574909
Epoch 40, change: 0.0031236228
Epoch 41, change: 0.0030883713
```

Epoch 42, change: 0.0030550848
Epoch 43, change: 0.0030229243
Epoch 44, change: 0.002993819
Epoch 45, change: 0.0029638736
Epoch 46, change: 0.002934612
Epoch 47, change: 0.002905708
Epoch 48, change: 0.0028713694
Epoch 49, change: 0.0028437968
Epoch 50, change: 0.0028182828
Epoch 51, change: 0.0027902448
Epoch 52, change: 0.0027614847
Epoch 53, change: 0.0027377574
Epoch 54, change: 0.0027125761
Epoch 55, change: 0.0026816045
Epoch 56, change: 0.0026564785
Epoch 57, change: 0.0026335059
Epoch 58, change: 0.0026044306
Epoch 59, change: 0.0025797196
Epoch 60, change: 0.0025560841
Epoch 61, change: 0.0025269997
Epoch 62, change: 0.0025040453
Epoch 63, change: 0.002477797
Epoch 64, change: 0.0024525792
Epoch 65, change: 0.0024334008
Epoch 66, change: 0.0024127199
Epoch 67, change: 0.0023721863
Epoch 68, change: 0.0023592771
Epoch 69, change: 0.0023244738
Epoch 70, change: 0.0022994694
Epoch 71, change: 0.0022818762
Epoch 72, change: 0.0022448783
Epoch 73, change: 0.0022157695
Epoch 74, change: 0.002193344
Epoch 75, change: 0.0021762126
Epoch 76, change: 0.0021581855
Epoch 77, change: 0.0021020662
Epoch 78, change: 0.0020807166
Epoch 79, change: 0.0020701065
Epoch 80, change: 0.0020530312
Epoch 81, change: 0.0019909
Epoch 82, change: 0.0019795786
Epoch 83, change: 0.0019232372
Epoch 84, change: 0.0019004431
Epoch 85, change: 0.0018900938
Epoch 86, change: 0.0018749765
Epoch 87, change: 0.0018599486
Epoch 88, change: 0.0017527566
Epoch 89, change: 0.0017357631
Epoch 90, change: 0.0017073144
Epoch 91, change: 0.0016688194
Epoch 92, change: 0.0016275956
Epoch 93, change: 0.0016073553
Epoch 94, change: 0.0015452054
Epoch 95, change: 0.0015570332
Epoch 96, change: 0.0014659382
Epoch 97, change: 0.0014439234

Epoch 98, change: 0.001447049
Epoch 99, change: 0.001350488
Epoch 100, change: 0.0013231553
Epoch 101, change: 0.001274997
Epoch 102, change: 0.0012872203
Epoch 103, change: 0.0012771395
Epoch 104, change: 0.001125279
Epoch 105, change: 0.0011215315
Epoch 106, change: 0.0011173364
Epoch 107, change: 0.0010480806
Epoch 108, change: 0.001024544
Epoch 109, change: 0.0010321577
Epoch 110, change: 0.0010241368
Epoch 111, change: 0.0010160987
Epoch 112, change: 0.00075589894
Epoch 113, change: 0.00083048684
Epoch 114, change: 0.00078595677
Epoch 115, change: 0.00079311192
Epoch 116, change: 0.00071282149
Epoch 117, change: 0.00069726971
Epoch 118, change: 0.00069359865
Epoch 119, change: 0.00068821753
Epoch 120, change: 0.00057887571
Epoch 121, change: 0.00056722063
Epoch 122, change: 0.00057892483
Epoch 123, change: 0.0005744784
Epoch 124, change: 0.00047226324
Epoch 125, change: 0.0005037694
Epoch 126, change: 0.00049996368
Epoch 127, change: 0.00048292599
Epoch 128, change: 0.00043812741
Epoch 129, change: 0.00043490324
Epoch 130, change: 0.00043170328
Epoch 131, change: 0.00047985007
Epoch 132, change: 0.00045197508
Epoch 133, change: 0.00044289418
Epoch 134, change: 0.00047868259
Epoch 135, change: 0.00046519625
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Epoch 137, change: 0.00045880867
Epoch 138, change: 0.00045340771
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Epoch 140, change: 0.00042661284
Epoch 141, change: 0.00040923926
Epoch 142, change: 0.00040064041
Epoch 143, change: 0.00049736727
Epoch 144, change: 0.00044391989
Epoch 145, change: 0.00044435837
Epoch 146, change: 0.00042876891
Epoch 147, change: 0.00041570778
Epoch 148, change: 0.00042593116
Epoch 149, change: 0.0004183922
Epoch 150, change: 0.00040235348
Epoch 151, change: 0.00040886357
Epoch 152, change: 0.00039042972
Epoch 153, change: 0.00039388437

Epoch 154, change: 0.00038588503
Epoch 155, change: 0.00037889539
Epoch 156, change: 0.00036707787
Epoch 157, change: 0.00035806705
Epoch 158, change: 0.0003479704
Epoch 159, change: 0.00033999072
Epoch 160, change: 0.00036502373
Epoch 161, change: 0.00034565707
Epoch 162, change: 0.00033934257
Epoch 163, change: 0.00033405027
Epoch 164, change: 0.00032487435
Epoch 165, change: 0.00032018979
Epoch 166, change: 0.0003100965
Epoch 167, change: 0.00031037631
Epoch 168, change: 0.00030227122
Epoch 169, change: 0.00029379405
Epoch 170, change: 0.00028678025
Epoch 171, change: 0.00029002378
Epoch 172, change: 0.0002813286
Epoch 173, change: 0.00027496757
Epoch 174, change: 0.00026729267
Epoch 175, change: 0.00026586578
Epoch 176, change: 0.00025956011
Epoch 177, change: 0.00025330883
Epoch 178, change: 0.00024749063
Epoch 179, change: 0.0002458978
Epoch 180, change: 0.00023735908
Epoch 181, change: 0.00023614373
Epoch 182, change: 0.00022892998
Epoch 183, change: 0.00022364685
Epoch 184, change: 0.00022252363
Epoch 185, change: 0.00021570237
Epoch 186, change: 0.0002116828
Epoch 187, change: 0.00020689841
Epoch 188, change: 0.00020118075
Epoch 189, change: 0.00019934851
Epoch 190, change: 0.00019501831
Epoch 191, change: 0.00019017581
Epoch 192, change: 0.00018658733
Epoch 193, change: 0.00018238766
Epoch 194, change: 0.0001794742
Epoch 195, change: 0.00017529451
Epoch 196, change: 0.0001707341
Epoch 197, change: 0.00016815389
Epoch 198, change: 0.0001636659
Epoch 199, change: 0.00016092652
Epoch 200, change: 0.00015665841
Epoch 201, change: 0.00015512239
Epoch 202, change: 0.00015047327
Epoch 203, change: 0.00014828339
Epoch 204, change: 0.00014562872
Epoch 205, change: 0.00014386222
Epoch 206, change: 0.00014249567
Epoch 207, change: 0.00014129556
Epoch 208, change: 0.00013842093
Epoch 209, change: 0.00013596835

```
Epoch 210, change: 0.00013511352
Epoch 211, change: 0.00013338517
Epoch 212, change: 0.0001319486
Epoch 213, change: 0.00013025253
Epoch 214, change: 0.00012796131
Epoch 215, change: 0.00012627515
Epoch 216, change: 0.00012552725
Epoch 217, change: 0.00012381511
Epoch 218, change: 0.0001221831
Epoch 219, change: 0.00012012501
Epoch 220, change: 0.00011918862
Epoch 221, change: 0.0001175666
Epoch 222, change: 0.00011641687
Epoch 223, change: 0.00011487017
Epoch 224, change: 0.0001131049
Epoch 225, change: 0.00011184489
Epoch 226, change: 0.00011082138
Epoch 227, change: 0.00010925085
Epoch 228, change: 0.00010759726
Epoch 229, change: 0.00010679269
Epoch 230, change: 0.00010518406
Epoch 231, change: 0.00010461578
Epoch 232, change: 0.00010317537
Epoch 233, change: 0.00010160392
Epoch 234, change: 0.00010043816
convergence after 235 epochs took 229 seconds
```

Out[19]:

```
LogisticRegression
LogisticRegression(max_iter=500, multi_class='multinomial',
                    solver='saga',
                    verbose=1)
```

```
In [ ]: #better safe than sorry
np.save('X_train.npy', X_train)
np.save('X_test.npy', X_test)
np.save('y_train.npy', y_train)
np.save('y_test.npy', y_test)
```

Result evaluation

```
In [20]: from sklearn.metrics import classification_report

y_pred = model.predict(X_test)

print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
1.14.14.18	0.00	0.00	0.00	1
1.15.1.1	0.00	0.00	0.00	1
1.16.3.1	0.50	1.00	0.67	1
1.17.4.1	0.00	0.00	0.00	1
1.5.98.3	0.00	0.00	0.00	1

1.8.3.2	0.00	0.00	0.00	1
2.1.1.354	0.00	0.00	0.00	1
2.1.1.360	0.00	0.00	0.00	1
2.1.1.37	0.00	0.00	0.00	1
2.1.1.72	0.00	0.00	0.00	1
2.1.1.77	0.20	1.00	0.33	1
2.1.1.86	0.00	0.00	0.00	1
2.1.3.15	0.00	0.00	0.00	1
2.3.1.225	1.00	1.00	1.00	1
2.3.1.269	0.00	0.00	0.00	1
2.3.1.48	0.00	0.00	0.00	1
2.3.1.51	0.00	0.00	0.00	1
2.3.2.23	0.00	0.00	0.00	1
2.3.2.26	0.33	1.00	0.50	1
2.3.2.27	0.00	0.00	0.00	1
2.3.2.31	0.00	0.00	0.00	1
2.4.1.109	0.00	0.00	0.00	1
2.4.1.198	0.00	0.00	0.00	1
2.4.2.31	0.00	0.00	0.00	1
2.5.1.18	0.00	0.00	0.00	1
2.7.1.107	0.00	0.00	0.00	1
2.7.1.21	0.00	0.00	0.00	1
2.7.1.33	0.00	0.00	0.00	1
2.7.1.67	0.00	0.00	0.00	1
2.7.10.1	0.00	0.00	0.00	1
2.7.10.2	0.00	0.00	0.00	1
2.7.11.1	0.00	0.00	0.00	1
2.7.11.25	0.20	1.00	0.33	1
2.7.13.3	0.00	0.00	0.00	1
2.7.4.3	0.00	0.00	0.00	1
2.7.6.1	0.00	0.00	0.00	1
2.7.7.108	0.00	0.00	0.00	1
2.7.7.19	0.00	0.00	0.00	1
2.7.7.48	0.00	0.00	0.00	1
2.7.7.49	0.00	0.00	0.00	1
2.7.7.6	0.00	0.00	0.00	1
2.7.7.65	0.00	0.00	0.00	1
2.7.7.7	0.00	0.00	0.00	1
2.7.8.7	0.00	0.00	0.00	1
2.8.1.13	0.00	0.00	0.00	1
3.1.1.1	0.00	0.00	0.00	1
3.1.1.3	0.00	0.00	0.00	1
3.1.1.32	0.00	0.00	0.00	1
3.1.1.4	0.00	0.00	0.00	1
3.1.1.96	0.00	0.00	0.00	1
3.1.11.2	0.00	0.00	0.00	1
3.1.11.6	1.00	1.00	1.00	1
3.1.12.1	0.00	0.00	0.00	1
3.1.13.4	0.00	0.00	0.00	1
3.1.21.10	0.00	0.00	0.00	1
3.1.21.4	0.00	0.00	0.00	1
3.1.26.3	0.00	0.00	0.00	1
3.1.26.4	0.00	0.00	0.00	1
3.1.26.5	0.00	0.00	0.00	1
3.1.3.16	0.00	0.00	0.00	1
3.1.3.18	0.00	0.00	0.00	1

3.1.3.2	0.00	0.00	0.00	1
3.1.3.3	0.00	0.00	0.00	1
3.1.3.48	0.00	0.00	0.00	1
3.1.3.5	0.00	0.00	0.00	1
3.1.4.12	0.00	0.00	0.00	1
3.1.4.35	0.00	0.00	0.00	1
3.1.4.4	0.00	0.00	0.00	1
3.1.4.52	0.00	0.00	0.00	1
3.2.1.1	0.00	0.00	0.00	1
3.2.1.14	0.00	0.00	0.00	1
3.2.1.17	0.00	0.00	0.00	1
3.2.1.18	0.00	0.00	0.00	1
3.2.1.22	0.00	0.00	0.00	1
3.2.1.23	0.00	0.00	0.00	1
3.2.1.35	0.00	0.00	0.00	1
3.2.1.39	0.00	0.00	0.00	1
3.2.1.4	0.00	0.00	0.00	1
3.2.1.52	0.00	0.00	0.00	1
3.2.1.55	0.00	0.00	0.00	1
3.2.1.78	0.00	0.00	0.00	1
3.2.1.8	1.00	1.00	1.00	1
3.2.1.96	0.00	0.00	0.00	1
3.2.2.22	0.00	0.00	0.00	1
3.2.2.6	0.00	0.00	0.00	1
3.4.19.12	0.00	0.00	0.00	1
3.4.21.105	0.00	0.00	0.00	1
3.4.22.49	0.00	0.00	0.00	1
3.4.23.36	0.00	0.00	0.00	1
3.5.1.28	0.00	0.00	0.00	1
3.5.1.98	0.00	0.00	0.00	1
3.5.2.6	0.00	0.00	0.00	1
3.6.1.1	0.00	0.00	0.00	1
3.6.1.13	0.00	0.00	0.00	1
3.6.1.23	0.00	0.00	0.00	1
3.6.1.5	0.00	0.00	0.00	1
3.6.1.9	0.00	0.00	0.00	1
3.6.4.12	0.00	0.00	0.00	1
3.6.4.13	0.00	0.00	0.00	1
3.8.1.2	0.50	1.00	0.67	1
4.2.1.1	0.00	0.00	0.00	1
4.2.1.10	0.00	0.00	0.00	1
4.2.1.2	0.00	0.00	0.00	1
4.2.1.20	0.33	1.00	0.50	1
4.2.2.2	0.00	0.00	0.00	1
4.2.2.3	0.00	0.00	0.00	1
4.2.2.n1	0.00	0.00	0.00	1
4.4.1.3	0.00	0.00	0.00	1
4.6.1.1	0.00	0.00	0.00	1
4.6.1.16	0.00	0.00	0.00	1
4.6.1.2	0.00	0.00	0.00	1
5.2.1.8	0.00	0.00	0.00	1
5.3.4.1	0.50	1.00	0.67	1
5.4.99.5	0.00	0.00	0.00	1
5.6.2.2	0.00	0.00	0.00	1
5.6.2.4	0.00	0.00	0.00	1
6.1.1.19	0.00	0.00	0.00	1

6.3.2.4	0.00	0.00	0.00	1
6.3.4.19	0.00	0.00	0.00	1
6.3.5.2	0.00	0.00	0.00	1
6.3.5.5	1.00	1.00	1.00	1
6.5.1.1	0.00	0.00	0.00	1
6.5.1.3	0.00	0.00	0.00	1
7.1.1.2	0.00	0.00	0.00	1
7.1.1.8	0.00	0.00	0.00	1
7.1.1.9	0.00	0.00	0.00	1
7.1.2.2	0.00	0.00	0.00	1
7.2.1.1	0.00	0.00	0.00	1
accuracy			0.09	128
macro avg	0.05	0.09	0.06	128
weighted avg	0.05	0.09	0.06	128

```

/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/s
klearn/metrics/_classification.py:1565: UndefinedMetricWarning: Prec
ision is ill-defined and being set to 0.0 in labels with no predicte
d samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(resu
lt))
/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/s
klearn/metrics/_classification.py:1565: UndefinedMetricWarning: Prec
ision is ill-defined and being set to 0.0 in labels with no predicte
d samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(resu
lt))
/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/s
klearn/metrics/_classification.py:1565: UndefinedMetricWarning: Prec
ision is ill-defined and being set to 0.0 in labels with no predicte
d samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(resu
lt))

```

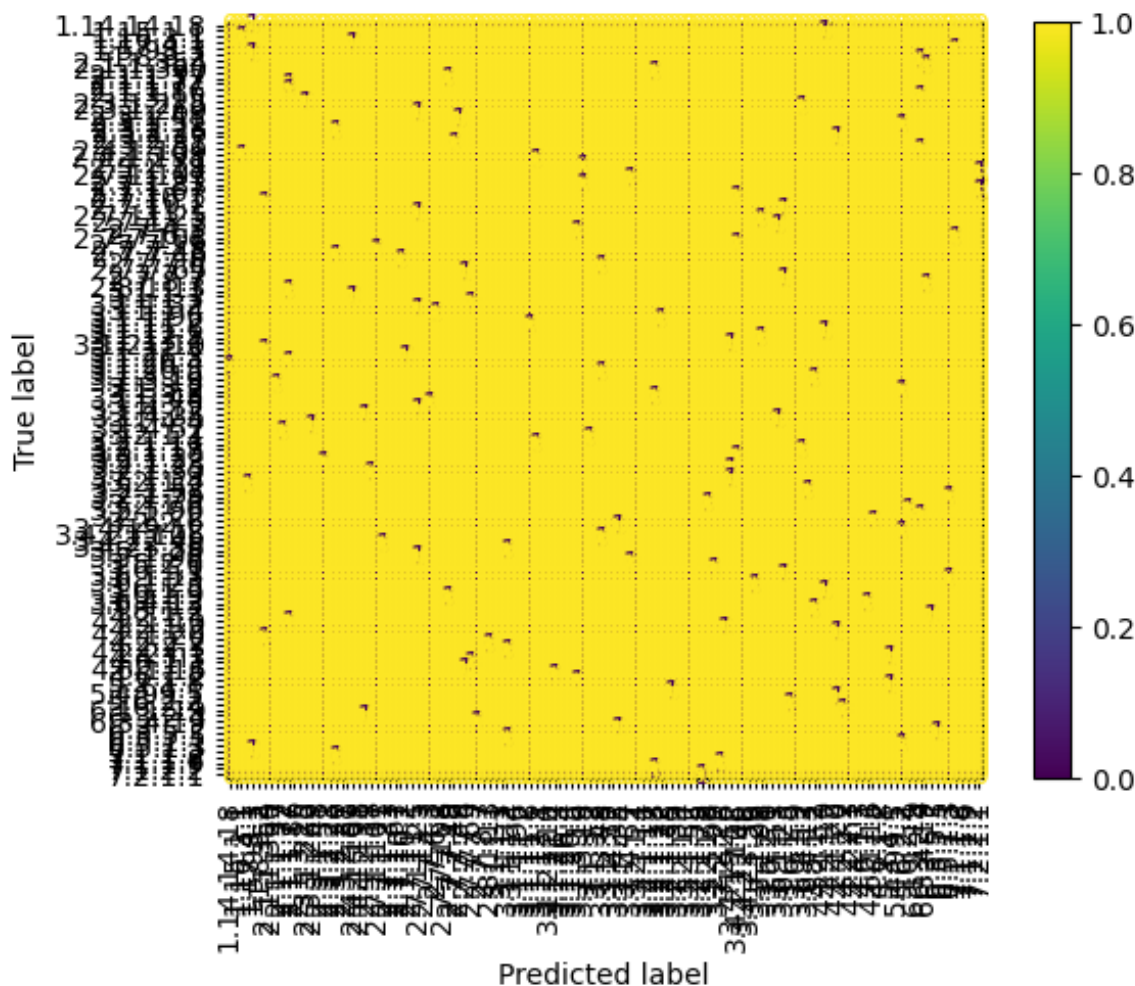
Test Model performance

```

In [ ]: import matplotlib.pyplot as plt
from sklearn.metrics import ConfusionMatrixDisplay

# Confusion Matrix
ConfusionMatrixDisplay.from_estimator(model, X_test, y_test)
plt.xticks(rotation=90)
plt.show()

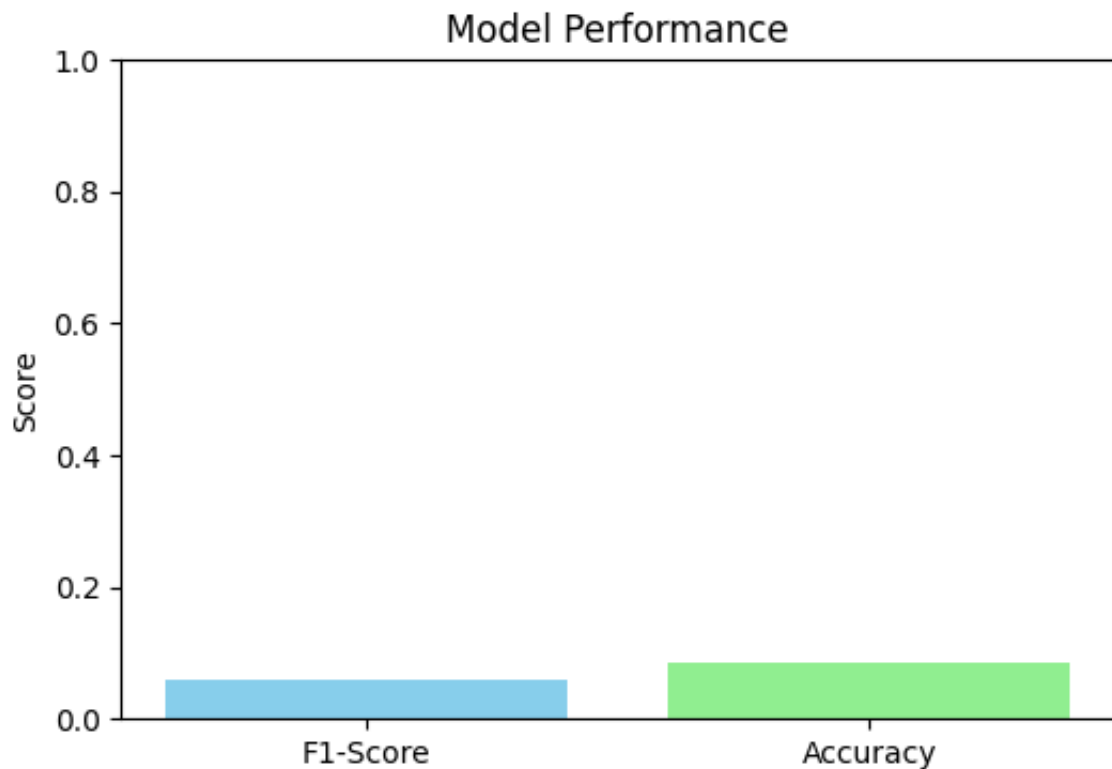
```



```
In [ ]: # Result-Plot: F1-Score und Accuracy
f1 = f1_score(y_test, y_pred, average='weighted')
acc = accuracy_score(y_test, y_pred)

plt.figure(figsize=(6,4))
plt.bar(['F1-Score', 'Accuracy'], [f1, acc], color=['skyblue', 'lightcoral'])
plt.ylim(0,1)
plt.title('Model Performance')
plt.ylabel('Score')
plt.show()

print("F1-Score:", f1)
print("Accuracy:", acc)
```



F1-Score: 0.05989583333333333

Accuracy: 0.0859375

k-mer method increases the result from F1-score of 0.057 and accuracy of 8.5% to F1-score 0.1513 and accuracy = 19.53%

```
In [ ]: # Save model
import joblib
joblib.dump(model, 'ec_model.joblib')
print("Model saved as ec_model.joblib")
```

Model saved as ec_model.joblib

Train logistic regression models with different values of regularization strength (C)

Train model with different C values

```
In [12]: # Define values for C (inverse of regularization strength)
C_values = [0.01, 0.1, 1, 10, 100]
accuracy_scores = []
f1_scores = []

for C in C_values:
    model = LogisticRegression(
        solver='saga',
        multi_class='multinomial',
        max_iter=300,
```

```

C=C,
penalty='l2',
verbose=0
)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)

acc = accuracy_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred, average='weighted')

accuracy_scores.append(acc)
f1_scores.append(f1)
print(f"C = {C:<6} → Accuracy = {acc:.4f}, F1-Score = {f1:.4f}")

```

/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/sklearn/linear_model/_logistic.py:1247: FutureWarning: 'multi_class' was deprecated in version 1.5 and will be removed in 1.7. From then on, it will always use 'multinomial'. Leave it to its default value to avoid this warning.

warnings.warn(

C = 0.01 → Accuracy = 0.0547, F1-Score = 0.0283

/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/sklearn/linear_model/_logistic.py:1247: FutureWarning: 'multi_class' was deprecated in version 1.5 and will be removed in 1.7. From then on, it will always use 'multinomial'. Leave it to its default value to avoid this warning.

warnings.warn(

/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/sklearn/linear_model/_sag.py:348: ConvergenceWarning: The max_iter was reached which means the coef_ did not converge

warnings.warn(

/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/sklearn/linear_model/_logistic.py:1247: FutureWarning: 'multi_class' was deprecated in version 1.5 and will be removed in 1.7. From then on, it will always use 'multinomial'. Leave it to its default value to avoid this warning.

warnings.warn(

C = 0.1 → Accuracy = 0.0859, F1-Score = 0.0599

C = 1 → Accuracy = 0.0859, F1-Score = 0.0599

/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/sklearn/linear_model/_logistic.py:1247: FutureWarning: 'multi_class' was deprecated in version 1.5 and will be removed in 1.7. From then on, it will always use 'multinomial'. Leave it to its default value to avoid this warning.

warnings.warn(

/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/sklearn/linear_model/_sag.py:348: ConvergenceWarning: The max_iter was reached which means the coef_ did not converge

warnings.warn(

/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/sklearn/linear_model/_logistic.py:1247: FutureWarning: 'multi_class' was deprecated in version 1.5 and will be removed in 1.7. From then on, it will always use 'multinomial'. Leave it to its default value to avoid this warning.

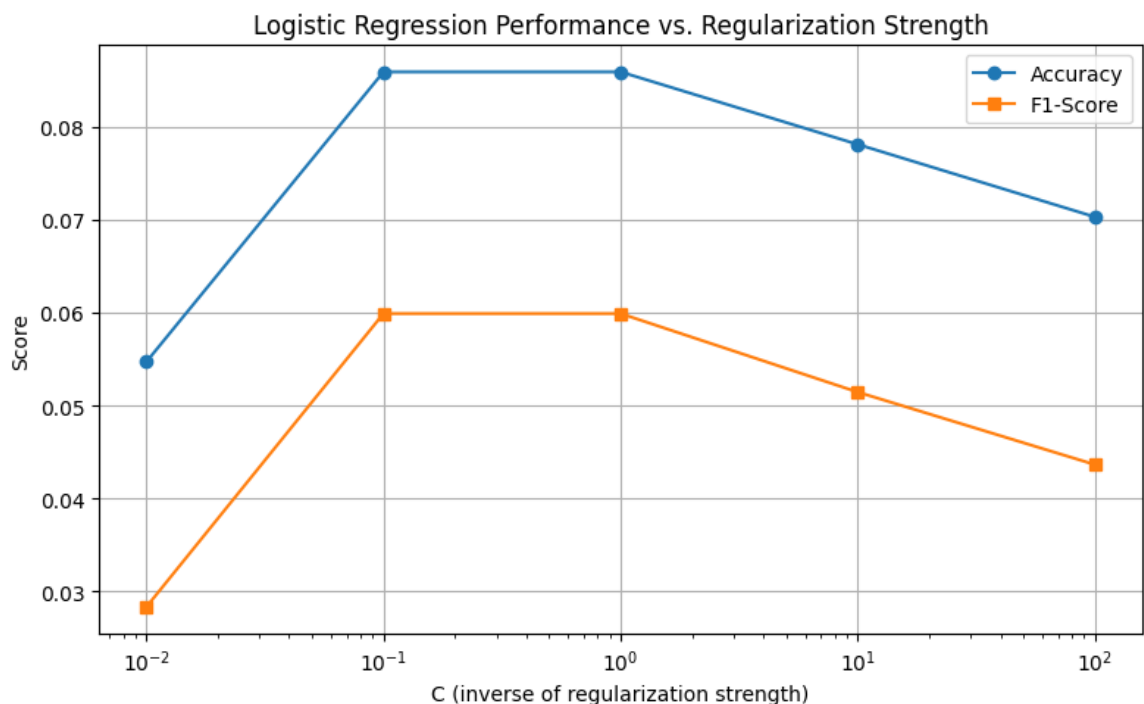
warnings.warn(

C = 10 → Accuracy = 0.0781, F1-Score = 0.0514
 C = 100 → Accuracy = 0.0703, F1-Score = 0.0436

```
/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/sklearn/linear_model/_sag.py:348: ConvergenceWarning: The max_iter was reached which means the coef_ did not converge
warnings.warn(
```

Evaluation

```
In [13]: plt.figure(figsize=(8, 5))
plt.plot(C_values, accuracy_scores, marker='o', label='Accuracy')
plt.plot(C_values, f1_scores, marker='s', label='F1-Score')
plt.xscale('log') # log scale is common for regularization
plt.xlabel('C (inverse of regularization strength)')
plt.ylabel('Score')
plt.title('Logistic Regression Performance vs. Regularization Strength')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```



Trying other encoding methods

k-mer frequencies

```
In [ ]: from sklearn.feature_extraction.text import CountVectorizer

# Prepare sequences as strings
sequences = [x['Sequence'] for x in train_data] # from your hugging
```

```

# Define k
k = 3 # 3-mer

# Create k-mer tokens from sequence
def seq_to_kmers(seq, k):
    return ' '.join([seq[i:i+k] for i in range(len(seq)-k+1)])

kmers = [seq_to_kmers(seq, k) for seq in sequences]

# Count frequency of each k-mer
vectorizer = CountVectorizer()
X_train = vectorizer.fit_transform(kmers)

print("Shape of Feature Matrix:", X_train.shape)

```

Shape of Feature Matrix: (512, 7986)

```

In [72]: sequences_test = [x['Sequence'] for x in test_data]
kmers_test = [seq_to_kmers(seq, k) for seq in sequences_test]
X_test = vectorizer.transform(kmers_test)

```

Logistic Regression

Setup and train model

```

In [ ]: from sklearn.linear_model import LogisticRegression

model = LogisticRegression(max_iter=500, solver='saga', multi_class=
# Klassifikationsmodell trainieren
model.fit(X_train, y_train)

```

/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/sklearn/linear_model/_logistic.py:1247: FutureWarning: 'multi_class' was deprecated in version 1.5 and will be removed in 1.7. From then on, it will always use 'multinomial'. Leave it to its default value to avoid this warning.

```

warnings.warn(
Epoch 1, change: 1
Epoch 2, change: 0.45378242
Epoch 3, change: 0.27358075
Epoch 4, change: 0.22511615
Epoch 5, change: 0.14311151
Epoch 6, change: 0.12305695
Epoch 7, change: 0.11305456
Epoch 8, change: 0.062747186
Epoch 9, change: 0.055638912
Epoch 10, change: 0.048084126
Epoch 11, change: 0.056927255
Epoch 12, change: 0.027057044
Epoch 13, change: 0.024994897
Epoch 14, change: 0.01809176
Epoch 15, change: 0.019658634
Epoch 16, change: 0.014534352
Epoch 17, change: 0.014924772

```

Epoch 18, change: 0.011722676
Epoch 19, change: 0.012346263
Epoch 20, change: 0.011838062
Epoch 21, change: 0.011191884
Epoch 22, change: 0.0091671033
Epoch 23, change: 0.0079999822
Epoch 24, change: 0.0083330926
Epoch 25, change: 0.00753895
Epoch 26, change: 0.0067818045
Epoch 27, change: 0.0070514842
Epoch 28, change: 0.0061054149
Epoch 29, change: 0.0061729374
Epoch 30, change: 0.005130764
Epoch 31, change: 0.0047472789
Epoch 32, change: 0.004981629
Epoch 33, change: 0.0047865694
Epoch 34, change: 0.0042141779
Epoch 35, change: 0.0042586396
Epoch 36, change: 0.0041635018
Epoch 37, change: 0.0030344354
Epoch 38, change: 0.0032561462
Epoch 39, change: 0.0030574572
Epoch 40, change: 0.0030396331
Epoch 41, change: 0.0029212716
Epoch 42, change: 0.0028943821
Epoch 43, change: 0.0028681866
Epoch 44, change: 0.0028444931
Epoch 45, change: 0.0028180697
Epoch 46, change: 0.002786203
Epoch 47, change: 0.0027654955
Epoch 48, change: 0.0027374997
Epoch 49, change: 0.0027117481
Epoch 50, change: 0.0026882816
Epoch 51, change: 0.0026618302
Epoch 52, change: 0.0026383427
Epoch 53, change: 0.002614577
Epoch 54, change: 0.0025880153
Epoch 55, change: 0.0025616788
Epoch 56, change: 0.0025425612
Epoch 57, change: 0.002512142
Epoch 58, change: 0.0024856638
Epoch 59, change: 0.0024579929
Epoch 60, change: 0.0024378442
Epoch 61, change: 0.0023999086
Epoch 62, change: 0.0023751888
Epoch 63, change: 0.0023523073
Epoch 64, change: 0.0023181191
Epoch 65, change: 0.0023027989
Epoch 66, change: 0.0022600706
Epoch 67, change: 0.0022335805
Epoch 68, change: 0.0022094618
Epoch 69, change: 0.0021925737
Epoch 70, change: 0.0021742633
Epoch 71, change: 0.0021072977
Epoch 72, change: 0.0020845041
Epoch 73, change: 0.0020590921

Epoch 74, change: 0.0020481654
Epoch 75, change: 0.001983954
Epoch 76, change: 0.001988619
Epoch 77, change: 0.0019194538
Epoch 78, change: 0.0018943445
Epoch 79, change: 0.001868623
Epoch 80, change: 0.0018150567
Epoch 81, change: 0.0018197971
Epoch 82, change: 0.0017480246
Epoch 83, change: 0.0017208579
Epoch 84, change: 0.001720924
Epoch 85, change: 0.0016327241
Epoch 86, change: 0.0016501804
Epoch 87, change: 0.0015364301
Epoch 88, change: 0.0015357605
Epoch 89, change: 0.0014947626
Epoch 90, change: 0.0014751967
Epoch 91, change: 0.001414682
Epoch 92, change: 0.0013773779
Epoch 93, change: 0.0013361726
Epoch 94, change: 0.0012972367
Epoch 95, change: 0.0012627049
Epoch 96, change: 0.0012177157
Epoch 97, change: 0.001178285
Epoch 98, change: 0.0011369337
Epoch 99, change: 0.0011398337
Epoch 100, change: 0.0011315073
Epoch 101, change: 0.00098288875
Epoch 102, change: 0.0009825601
Epoch 103, change: 0.000985533
Epoch 104, change: 0.00097844962
Epoch 105, change: 0.00097059348
Epoch 106, change: 0.00096583822
Epoch 107, change: 0.0009582009
Epoch 108, change: 0.00095165442
Epoch 109, change: 0.00094470025
Epoch 110, change: 0.00059324494
Epoch 111, change: 0.00066748648
Epoch 112, change: 0.00061540316
Epoch 113, change: 0.00063652218
Epoch 114, change: 0.00063053057
Epoch 115, change: 0.00053035803
Epoch 116, change: 0.00051578439
Epoch 117, change: 0.00051637598
Epoch 118, change: 0.00051199323
Epoch 119, change: 0.00048085705
Epoch 120, change: 0.0004761373
Epoch 121, change: 0.00055216776
Epoch 122, change: 0.00051607383
Epoch 123, change: 0.00048891321
Epoch 124, change: 0.00047334531
Epoch 125, change: 0.00054896535
Epoch 126, change: 0.00051771831
Epoch 127, change: 0.00051549516
Epoch 128, change: 0.00049420882
Epoch 129, change: 0.00048286706

Epoch 130, change: 0.00052132395
Epoch 131, change: 0.00048694926
Epoch 132, change: 0.0005075793
Epoch 133, change: 0.00048115161
Epoch 134, change: 0.00048861494
Epoch 135, change: 0.00047429031
Epoch 136, change: 0.0004830095
Epoch 137, change: 0.00046967852
Epoch 138, change: 0.00046735378
Epoch 139, change: 0.00044703544
Epoch 140, change: 0.00046096877
Epoch 141, change: 0.00044852591
Epoch 142, change: 0.0004383368
Epoch 143, change: 0.00042650145
Epoch 144, change: 0.00041466951
Epoch 145, change: 0.00040325988
Epoch 146, change: 0.00043068287
Epoch 147, change: 0.00041078004
Epoch 148, change: 0.00040376516
Epoch 149, change: 0.00039592618
Epoch 150, change: 0.00039000856
Epoch 151, change: 0.00038135194
Epoch 152, change: 0.00037441922
Epoch 153, change: 0.00036998181
Epoch 154, change: 0.0003626737
Epoch 155, change: 0.00035520998
Epoch 156, change: 0.00034543392
Epoch 157, change: 0.00034354666
Epoch 158, change: 0.00033361292
Epoch 159, change: 0.00033189957
Epoch 160, change: 0.0003238217
Epoch 161, change: 0.00031859875
Epoch 162, change: 0.00031204076
Epoch 163, change: 0.00030556863
Epoch 164, change: 0.00030013944
Epoch 165, change: 0.00029413281
Epoch 166, change: 0.00028621843
Epoch 167, change: 0.00028272383
Epoch 168, change: 0.00027665691
Epoch 169, change: 0.0002715644
Epoch 170, change: 0.00026648825
Epoch 171, change: 0.00025982526
Epoch 172, change: 0.00025317413
Epoch 173, change: 0.0002502447
Epoch 174, change: 0.00024421226
Epoch 175, change: 0.00023950191
Epoch 176, change: 0.00023544245
Epoch 177, change: 0.00023064627
Epoch 178, change: 0.00022724021
Epoch 179, change: 0.00022173197
Epoch 180, change: 0.00021582088
Epoch 181, change: 0.00021266336
Epoch 182, change: 0.00020763996
Epoch 183, change: 0.00020284878
Epoch 184, change: 0.00019983236
Epoch 185, change: 0.00019448764

```
Epoch 186, change: 0.0001909344
Epoch 187, change: 0.00018641355
Epoch 188, change: 0.00018147517
Epoch 189, change: 0.00017846594
Epoch 190, change: 0.00017621769
Epoch 191, change: 0.00017094119
Epoch 192, change: 0.00016838766
Epoch 193, change: 0.00016483753
Epoch 194, change: 0.0001603345
Epoch 195, change: 0.00015753454
Epoch 196, change: 0.00015302924
Epoch 197, change: 0.00014991824
Epoch 198, change: 0.00014736278
Epoch 199, change: 0.00014455289
Epoch 200, change: 0.00014072722
Epoch 201, change: 0.00013711497
Epoch 202, change: 0.00013498184
Epoch 203, change: 0.00013227701
Epoch 204, change: 0.00012867717
Epoch 205, change: 0.00012523158
Epoch 206, change: 0.0001226114
Epoch 207, change: 0.00011972365
Epoch 208, change: 0.00011946361
Epoch 209, change: 0.00011501576
Epoch 210, change: 0.00011237413
Epoch 211, change: 0.00011055124
Epoch 212, change: 0.00010716895
Epoch 213, change: 0.00010601603
Epoch 214, change: 0.00010269322
Epoch 215, change: 0.00010059382
convergence after 216 epochs took 209 seconds
```

Out[]:

```
LogisticRegression
LogisticRegression(max_iter=500, multi_class='multinomial',
                    solver='saga',
                    verbose=1)
```

In [50]:

```
#better safe than sorry
np.save('X_train.npy', X_train)
np.save('X_test.npy', X_test)
np.save('y_train.npy', y_train)
np.save('y_test.npy', y_test)
```

Result evaluation

In []:

```
from sklearn.metrics import classification_report

y_pred = model.predict(X_test)

print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
1.14.14.18	0.00	0.00	0.00	1
1.15.1.1	0.00	0.00	0.00	1
1.16.3.1	0.00	0.00	0.00	1
1.17.4.1	0.00	0.00	0.00	1
1.5.98.3	0.00	0.00	0.00	1
1.8.3.2	0.00	0.00	0.00	1
2.1.1.354	0.00	0.00	0.00	1
2.1.1.360	0.00	0.00	0.00	1
2.1.1.37	1.00	1.00	1.00	1
2.1.1.72	0.00	0.00	0.00	1
2.1.1.77	0.33	1.00	0.50	1
2.1.1.86	1.00	1.00	1.00	1
2.1.3.15	0.00	0.00	0.00	1
2.3.1.225	0.00	0.00	0.00	1
2.3.1.269	0.33	1.00	0.50	1
2.3.1.48	0.00	0.00	0.00	1
2.3.1.51	0.50	1.00	0.67	1
2.3.2.23	1.00	1.00	1.00	1
2.3.2.26	0.00	0.00	0.00	1
2.3.2.27	0.00	0.00	0.00	1
2.3.2.31	0.00	0.00	0.00	1
2.4.1.109	0.00	0.00	0.00	1
2.4.1.198	0.00	0.00	0.00	1
2.4.2.31	0.00	0.00	0.00	1
2.5.1.18	0.00	0.00	0.00	1
2.7.1.107	0.00	0.00	0.00	1
2.7.1.21	0.00	0.00	0.00	1
2.7.1.33	0.50	1.00	0.67	1
2.7.1.67	0.00	0.00	0.00	1
2.7.10.1	0.00	0.00	0.00	1
2.7.10.2	0.00	0.00	0.00	1
2.7.11.1	0.00	0.00	0.00	1
2.7.11.25	0.00	0.00	0.00	1
2.7.13.3	0.00	0.00	0.00	1
2.7.4.3	0.25	1.00	0.40	1
2.7.6.1	1.00	1.00	1.00	1
2.7.7.108	1.00	1.00	1.00	1
2.7.7.19	0.00	0.00	0.00	1
2.7.7.48	0.00	0.00	0.00	1
2.7.7.49	0.00	0.00	0.00	1
2.7.7.6	0.00	0.00	0.00	1
2.7.7.65	0.00	0.00	0.00	1
2.7.7.7	0.50	1.00	0.67	1
2.7.8.7	1.00	1.00	1.00	1
2.8.1.13	1.00	1.00	1.00	1
3.1.1.1	0.00	0.00	0.00	1
3.1.1.3	0.00	0.00	0.00	1
3.1.1.32	0.00	0.00	0.00	1
3.1.1.4	0.00	0.00	0.00	1
3.1.1.96	0.00	0.00	0.00	1
3.1.11.2	0.00	0.00	0.00	1
3.1.11.6	0.00	0.00	0.00	1
3.1.12.1	0.33	1.00	0.50	1
3.1.13.4	0.00	0.00	0.00	1

3.1.21.10	0.00	0.00	0.00	1
3.1.21.4	0.00	0.00	0.00	1
3.1.26.3	0.33	1.00	0.50	1
3.1.26.4	0.00	0.00	0.00	1
3.1.26.5	0.00	0.00	0.00	1
3.1.3.16	0.00	0.00	0.00	1
3.1.3.18	0.50	1.00	0.67	1
3.1.3.2	1.00	1.00	1.00	1
3.1.3.3	0.00	0.00	0.00	1
3.1.3.48	0.00	0.00	0.00	1
3.1.3.5	0.00	0.00	0.00	1
3.1.4.12	0.00	0.00	0.00	1
3.1.4.35	0.00	0.00	0.00	1
3.1.4.4	0.00	0.00	0.00	1
3.1.4.52	0.00	0.00	0.00	1
3.2.1.1	0.00	0.00	0.00	1
3.2.1.14	0.00	0.00	0.00	1
3.2.1.17	0.00	0.00	0.00	1
3.2.1.18	0.00	0.00	0.00	1
3.2.1.22	0.00	0.00	0.00	1
3.2.1.23	0.00	0.00	0.00	1
3.2.1.35	0.00	0.00	0.00	1
3.2.1.39	0.00	0.00	0.00	1
3.2.1.4	0.00	0.00	0.00	1
3.2.1.52	0.00	0.00	0.00	1
3.2.1.55	0.00	0.00	0.00	1
3.2.1.78	0.00	0.00	0.00	1
3.2.1.8	1.00	1.00	1.00	1
3.2.1.96	0.00	0.00	0.00	1
3.2.2.22	0.00	0.00	0.00	1
3.2.2.6	0.00	0.00	0.00	1
3.4.19.12	0.00	0.00	0.00	1
3.4.21.105	0.00	0.00	0.00	1
3.4.22.49	0.00	0.00	0.00	1
3.4.23.36	0.25	1.00	0.40	1
3.5.1.28	0.00	0.00	0.00	1
3.5.1.98	0.00	0.00	0.00	1
3.5.2.6	0.00	0.00	0.00	1
3.6.1.1	0.00	0.00	0.00	1
3.6.1.13	0.33	1.00	0.50	1
3.6.1.23	0.00	0.00	0.00	1
3.6.1.5	0.00	0.00	0.00	1
3.6.1.9	0.00	0.00	0.00	1
3.6.4.12	0.00	0.00	0.00	1
3.6.4.13	0.00	0.00	0.00	1
3.8.1.2	1.00	1.00	1.00	1
4.2.1.1	0.00	0.00	0.00	1
4.2.1.10	0.00	0.00	0.00	1
4.2.1.2	0.00	0.00	0.00	1
4.2.1.20	0.00	0.00	0.00	1
4.2.2.2	0.00	0.00	0.00	1
4.2.2.3	0.00	0.00	0.00	1
4.2.2.n1	1.00	1.00	1.00	1
4.4.1.3	0.00	0.00	0.00	1
4.6.1.1	0.00	0.00	0.00	1
4.6.1.16	0.00	0.00	0.00	1

4.6.1.2	0.00	0.00	0.00	1
5.2.1.8	0.00	0.00	0.00	1
5.3.4.1	0.00	0.00	0.00	1
5.4.99.5	0.00	0.00	0.00	1
5.6.2.2	0.00	0.00	0.00	1
5.6.2.4	0.00	0.00	0.00	1
6.1.1.19	0.00	0.00	0.00	1
6.3.2.4	0.25	1.00	0.40	1
6.3.4.19	0.00	0.00	0.00	1
6.3.5.2	1.00	1.00	1.00	1
6.3.5.5	1.00	1.00	1.00	1
6.5.1.1	0.00	0.00	0.00	1
6.5.1.3	0.00	0.00	0.00	1
7.1.1.2	0.00	0.00	0.00	1
7.1.1.8	0.00	0.00	0.00	1
7.1.1.9	0.00	0.00	0.00	1
7.1.2.2	0.00	0.00	0.00	1
7.2.1.1	0.00	0.00	0.00	1
accuracy				0.20
macro avg				0.14
weighted avg				0.14

```

/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/s
klearn/metrics/_classification.py:1565: UndefinedMetricWarning: Prec
ision is ill-defined and being set to 0.0 in labels with no predicte
d samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(resu
lt))
/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/s
klearn/metrics/_classification.py:1565: UndefinedMetricWarning: Prec
ision is ill-defined and being set to 0.0 in labels with no predicte
d samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(resu
lt))
/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/s
klearn/metrics/_classification.py:1565: UndefinedMetricWarning: Prec
ision is ill-defined and being set to 0.0 in labels with no predicte
d samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(resu
lt))

```

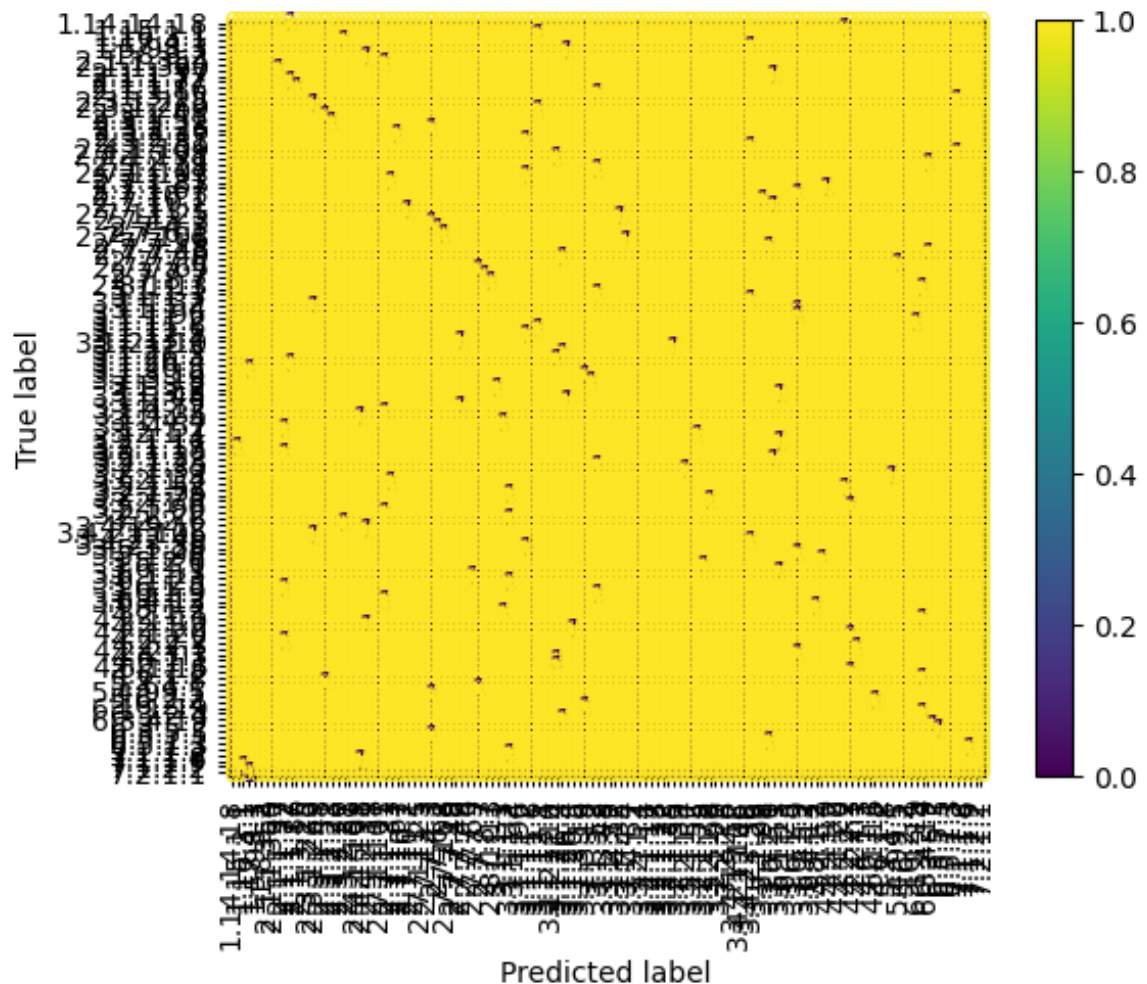
Test Model performance

```

In [75]: import matplotlib.pyplot as plt
from sklearn.metrics import ConfusionMatrixDisplay

# 9. Confusion Matrix
ConfusionMatrixDisplay.from_estimator(model, X_test, y_test)
plt.xticks(rotation=90)
plt.show()

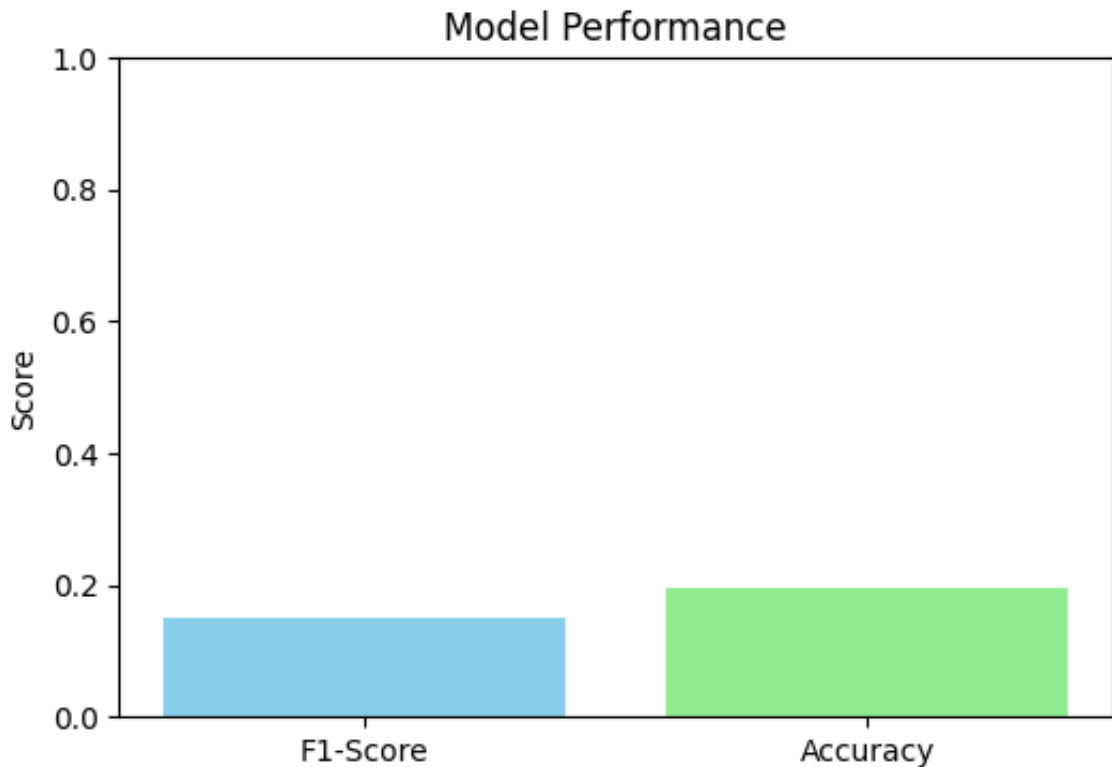
```



```
In [ ]: # Result-Plot: F1-Score und Accuracy
f1 = f1_score(y_test, y_pred, average='weighted')
acc = accuracy_score(y_test, y_pred)

plt.figure(figsize=(6,4))
plt.bar(['F1-Score', 'Accuracy'], [f1, acc], color=['skyblue', 'lightcoral'])
plt.ylim(0,1)
plt.title('Model Performance')
plt.ylabel('Score')
plt.show()

print("F1-Score:", f1)
print("Accuracy:", acc)
```



F1-Score: 0.15130208333333334

Accuracy: 0.1953125

k-mer method increases the result from F1-score of 0.057 and accuracy of 8.5%. to F1-score 0.1513 and accuracy = 19.53%

```
In [ ]: # Save model
import joblib
joblib.dump(model, 'ec_model.joblib')
print("Model saved as ec_model.joblib")
```

Model saved as ec_model.joblib

Comparison with the models from DGEBSM2

```
In [7]: #! pip install dgeb
! pip install transformers datasets torch
```

/Users/jonathanwolpert/.zshenv::1: no such file or directory: /Users/jonathanwolpert/.cargo/env

Defaulting to user installation because normal site-packages is not writeable

WARNING: Ignoring invalid distribution -cikit-learn (/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages)

WARNING: Ignoring invalid distribution -cikit-learn (/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages)

Requirement already satisfied: transformers in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (4.51.1)

Requirement already satisfied: datasets in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (3.5.0)

Requirement already satisfied: torch in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (2.6.0)

Requirement already satisfied: filelock in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from transformers) (3.18.0)

Requirement already satisfied: huggingface-hub<1.0,>=0.30.0 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from transformers) (0.30.2)

Requirement already satisfied: numpy>=1.17 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from transformers) (2.0.2)

Requirement already satisfied: packaging>=20.0 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from transformers) (24.0)

Requirement already satisfied: pyyaml>=5.1 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from transformers) (6.0.2)

Requirement already satisfied: regex!=2019.12.17 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from transformers) (2024.11.6)

Requirement already satisfied: requests in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from transformers) (2.32.3)

Requirement already satisfied: tokenizers<0.22,>=0.21 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from transformers) (0.21.1)

Requirement already satisfied: safetensors>=0.4.3 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from transformers) (0.5.3)

Requirement already satisfied: tqdm>=4.27 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from transformers) (4.67.1)

Requirement already satisfied: pyarrow>=15.0.0 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from datasets) (19.0.1)

Requirement already satisfied: dill<0.3.9,>=0.3.0 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from datasets) (0.3.8)

Requirement already satisfied: pandas in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from datasets) (2.2.2)

Requirement already satisfied: xxhash in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from datasets) (3.5.0)

Requirement already satisfied: multiprocess<0.70.17 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from datasets) (0.70.16)

Requirement already satisfied: fsspec<=2024.12.0,>=2023.1.0 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from fsspec[http]<=2024.12.0,>=2023.1.0->datasets) (2024.12.0)

Requirement already satisfied: aiohttp in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from datasets) (3.11.16)

Requirement already satisfied: typing-extensions>=4.10.0 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from torch) (4.13.1)

Requirement already satisfied: networkx in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from torch) (3.2.1)

Requirement already satisfied: jinja2 in /Users/jonathanwolpert/Libr

ary/Python/3.9/lib/python/site-packages (from torch) (3.1.5)
Requirement already satisfied: sympy==1.13.1 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from torch) (1.13.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from sympy==1.13.1->torch) (1.3.0)
Requirement already satisfied: aiohappyeyeballs>=2.3.0 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from aiohttp->datasets) (2.6.1)
Requirement already satisfied: aiosignal>=1.1.2 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from aiohttp->datasets) (1.3.2)
Requirement already satisfied: async-timeout<6.0,>=4.0 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from aiohttp->datasets) (5.0.1)
Requirement already satisfied: attrs>=17.3.0 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from aiohttp->datasets) (25.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from aiohttp->datasets) (1.5.0)
Requirement already satisfied: multidict<7.0,>=4.5 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from aiohttp->datasets) (6.2.0)
Requirement already satisfied: propcache>=0.2.0 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from aiohttp->datasets) (0.3.1)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from aiohttp->datasets) (1.19.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from requests->transformers) (3.4.1)
Requirement already satisfied: idna<4,>=2.5 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from requests->transformers) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from requests->transformers) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from requests->transformers) (2025.1.31)
Requirement already satisfied: MarkupSafe>=2.0 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from jinja2->torch) (3.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from pandas->datasets) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from pandas->datasets) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in /Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages (from pandas->datasets) (2024.1)
Requirement already satisfied: six>=1.5 in /Applications/Xcode.app/Contents/Developer/Library/Frameworks/Python3.framework/Versions/3.9/

```
lib/python3.9/site-packages (from python-dateutil>=2.8.2->pandas->datasets) (1.15.0)
```

```
WARNING: Ignoring invalid distribution -cikit-learn (/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages)
```

```
WARNING: Ignoring invalid distribution -cikit-learn (/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages)
```

Load Dataset

```
In [1]: from datasets import load_dataset
```

```
dataset = load_dataset("tattabio/ec_classification")
train_data = dataset["train"]
test_data = dataset["test"]
```

```
/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/urllib3/__init__.py:35: NotOpenSSLWarning: urllib3 v2 only supports 0
penSSL 1.1.1+, currently the 'ssl' module is compiled with 'LibreSSL
2.8.3'. See: https://github.com/urllib3/urllib3/issues/3020
```

```
warnings.warn(
```

```
/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/tqdm/auto.py:21: TqdmWarning: IProgress not found. Please update jupyter
ter and ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/
user_install.html
```

```
from .autonotebook import tqdm as notebook_tqdm
```

Load ESM2 model & tokenizer

```
In [4]: model_name = "facebook/esm2_t6_8M_UR50D"
```

```
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModel.from_pretrained(model_name)
model.eval() # disable training mode
```

```
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2_t6_8M_UR50D and are newly initialized: ['pooler.dense.bias', 'pooler.dense.weight']
```

```
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
```

```

Out[4]: EsmModel(
  (embeddings): EsmEmbeddings(
    (word_embeddings): Embedding(33, 320, padding_idx=1)
    (dropout): Dropout(p=0.0, inplace=False)
    (position_embeddings): Embedding(1026, 320, padding_idx=1)
  )
  (encoder): EsmEncoder(
    (layer): ModuleList(
      (0-5): 6 x EsmLayer(
        (attention): EsmAttention(
          (self): EsmSelfAttention(
            (query): Linear(in_features=320, out_features=320, bias=True)
            (key): Linear(in_features=320, out_features=320, bias=True)
            (value): Linear(in_features=320, out_features=320, bias=True)
            (dropout): Dropout(p=0.0, inplace=False)
            (rotary_embeddings): RotaryEmbedding()
          )
          (output): EsmSelfOutput(
            (dense): Linear(in_features=320, out_features=320, bias=True)
            (dropout): Dropout(p=0.0, inplace=False)
          )
          (LayerNorm): LayerNorm((320,), eps=1e-05, elementwise_affine=True)
        )
        (intermediate): EsmIntermediate(
          (dense): Linear(in_features=320, out_features=1280, bias=True)
        )
        (output): EsmOutput(
          (dense): Linear(in_features=1280, out_features=320, bias=True)
          (dropout): Dropout(p=0.0, inplace=False)
        )
        (LayerNorm): LayerNorm((320,), eps=1e-05, elementwise_affine=True)
      )
    )
    (emb_layer_norm_after): LayerNorm((320,), eps=1e-05, elementwise_affine=True)
  )
  (pooler): EsmPooler(
    (dense): Linear(in_features=320, out_features=320, bias=True)
    (activation): Tanh()
  )
  (contact_head): EsmContactPredictionHead(
    (regression): Linear(in_features=120, out_features=1, bias=True)
    (activation): Sigmoid()
  )
)

```

Define embedding function (mean pooling)

```
In [5]: def get_esm2_embedding(sequence):
        inputs = tokenizer(sequence, return_tensors="pt", truncation=Tr
        with torch.no_grad():
            outputs = model(**inputs)
        # Average pooling over token embeddings
        embedding = outputs.last_hidden_state.mean(dim=1)
        return embedding[0].numpy()
```

Apply to training & test sequences

DGEB Data Set

```
In [6]: # Convert huggingface datasets to lists
train_sequences = [x["Sequence"] for x in train_data]
train_labels = [x["Label"] for x in train_data]

test_sequences = [x["Sequence"] for x in test_data]
test_labels = [x["Label"] for x in test_data]

# Generate embeddings
X_train_embed = np.stack([get_esm2_embedding(seq) for seq in train_
X_test_embed = np.stack([get_esm2_embedding(seq) for seq in test_se
```

Asking to truncate to max_length but no maximum length is provided and the model has no predefined maximum length. Default to no truncation.

Train classifier

```
In [7]: model = LogisticRegression(max_iter=500, solver='saga', multi_class
model.fit(X_train_embed, train_labels)

y_pred = model.predict(X_test_embed)
print(classification_report(test_labels, y_pred))
```

/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/sklearn/linear_model/_logistic.py:1247: FutureWarning: 'multi_class' was deprecated in version 1.5 and will be removed in 1.7. From then on, it will always use 'multinomial'. Leave it to its default value to avoid this warning.

```
warnings.warn(
precision    recall  f1-score   support

1.14.14.18      0.00      0.00      0.00         1
1.15.1.1         1.00      1.00      1.00         1
1.16.3.1         0.00      0.00      0.00         1
1.17.4.1         1.00      1.00      1.00         1
1.5.98.3         0.00      0.00      0.00         1
1.8.3.2          0.00      0.00      0.00         1
```

2.1.1.354	0.00	0.00	0.00	1
2.1.1.360	0.00	0.00	0.00	1
2.1.1.37	0.50	1.00	0.67	1
2.1.1.72	0.00	0.00	0.00	1
2.1.1.77	1.00	1.00	1.00	1
2.1.1.86	0.50	1.00	0.67	1
2.1.3.15	0.00	0.00	0.00	1
2.3.1.225	0.00	0.00	0.00	1
2.3.1.269	1.00	1.00	1.00	1
2.3.1.48	0.00	0.00	0.00	1
2.3.1.51	1.00	1.00	1.00	1
2.3.2.23	0.00	0.00	0.00	1
2.3.2.26	0.17	1.00	0.29	1
2.3.2.27	0.00	0.00	0.00	1
2.3.2.31	0.00	0.00	0.00	1
2.4.1.109	1.00	1.00	1.00	1
2.4.1.198	0.33	1.00	0.50	1
2.4.2.31	0.00	0.00	0.00	1
2.5.1.18	1.00	1.00	1.00	1
2.7.1.107	0.00	0.00	0.00	1
2.7.1.21	0.00	0.00	0.00	1
2.7.1.33	0.50	1.00	0.67	1
2.7.1.67	0.00	0.00	0.00	1
2.7.10.1	0.50	1.00	0.67	1
2.7.10.2	0.00	0.00	0.00	1
2.7.11.1	0.00	0.00	0.00	1
2.7.11.25	0.00	0.00	0.00	1
2.7.13.3	0.00	0.00	0.00	1
2.7.4.3	1.00	1.00	1.00	1
2.7.6.1	1.00	1.00	1.00	1
2.7.7.108	0.50	1.00	0.67	1
2.7.7.19	0.00	0.00	0.00	1
2.7.7.48	1.00	1.00	1.00	1
2.7.7.49	1.00	1.00	1.00	1
2.7.7.6	0.00	0.00	0.00	1
2.7.7.65	0.00	0.00	0.00	1
2.7.7.7	0.00	0.00	0.00	1
2.7.8.7	0.00	0.00	0.00	1
2.8.1.13	1.00	1.00	1.00	1
3.1.1.1	1.00	1.00	1.00	1
3.1.1.3	1.00	1.00	1.00	1
3.1.1.32	1.00	1.00	1.00	1
3.1.1.4	0.00	0.00	0.00	1
3.1.1.96	0.00	0.00	0.00	1
3.1.11.2	0.00	0.00	0.00	1
3.1.11.6	0.50	1.00	0.67	1
3.1.12.1	1.00	1.00	1.00	1
3.1.13.4	0.00	0.00	0.00	1
3.1.21.10	0.00	0.00	0.00	1
3.1.21.4	0.00	0.00	0.00	1
3.1.26.3	0.00	0.00	0.00	1
3.1.26.4	1.00	1.00	1.00	1
3.1.26.5	0.00	0.00	0.00	1
3.1.3.16	1.00	1.00	1.00	1
3.1.3.18	0.50	1.00	0.67	1
3.1.3.2	0.00	0.00	0.00	1

3.1.3.3	1.00	1.00	1.00	1
3.1.3.48	1.00	1.00	1.00	1
3.1.3.5	0.00	0.00	0.00	1
3.1.4.12	0.00	0.00	0.00	1
3.1.4.35	1.00	1.00	1.00	1
3.1.4.4	0.00	0.00	0.00	1
3.1.4.52	1.00	1.00	1.00	1
3.2.1.1	0.00	0.00	0.00	1
3.2.1.14	0.00	0.00	0.00	1
3.2.1.17	0.00	0.00	0.00	1
3.2.1.18	1.00	1.00	1.00	1
3.2.1.22	0.50	1.00	0.67	1
3.2.1.23	0.00	0.00	0.00	1
3.2.1.35	0.00	0.00	0.00	1
3.2.1.39	0.00	0.00	0.00	1
3.2.1.4	0.00	0.00	0.00	1
3.2.1.52	1.00	1.00	1.00	1
3.2.1.55	0.00	0.00	0.00	1
3.2.1.78	1.00	1.00	1.00	1
3.2.1.8	1.00	1.00	1.00	1
3.2.1.96	0.00	0.00	0.00	1
3.2.2.22	0.00	0.00	0.00	1
3.2.2.6	0.00	0.00	0.00	1
3.4.19.12	0.00	0.00	0.00	1
3.4.21.105	0.00	0.00	0.00	1
3.4.22.49	0.00	0.00	0.00	1
3.4.23.36	1.00	1.00	1.00	1
3.5.1.28	1.00	1.00	1.00	1
3.5.1.98	0.33	1.00	0.50	1
3.5.2.6	1.00	1.00	1.00	1
3.6.1.1	0.00	0.00	0.00	1
3.6.1.13	1.00	1.00	1.00	1
3.6.1.23	0.00	0.00	0.00	1
3.6.1.5	1.00	1.00	1.00	1
3.6.1.9	0.00	0.00	0.00	1
3.6.4.12	0.00	0.00	0.00	1
3.6.4.13	1.00	1.00	1.00	1
3.8.1.2	1.00	1.00	1.00	1
4.2.1.1	0.00	0.00	0.00	1
4.2.1.10	0.33	1.00	0.50	1
4.2.1.2	0.50	1.00	0.67	1
4.2.1.20	0.00	0.00	0.00	1
4.2.2.2	0.00	0.00	0.00	1
4.2.2.3	0.50	1.00	0.67	1
4.2.2.n1	1.00	1.00	1.00	1
4.4.1.3	0.00	0.00	0.00	1
4.6.1.1	0.00	0.00	0.00	1
4.6.1.16	0.00	0.00	0.00	1
4.6.1.2	0.00	0.00	0.00	1
5.2.1.8	1.00	1.00	1.00	1
5.3.4.1	1.00	1.00	1.00	1
5.4.99.5	0.00	0.00	0.00	1
5.6.2.2	1.00	1.00	1.00	1
5.6.2.4	0.00	0.00	0.00	1
6.1.1.19	0.00	0.00	0.00	1
6.3.2.4	1.00	1.00	1.00	1

6.3.4.19	1.00	1.00	1.00	1
6.3.5.2	1.00	1.00	1.00	1
6.3.5.5	1.00	1.00	1.00	1
6.5.1.1	0.00	0.00	0.00	1
6.5.1.3	0.00	0.00	0.00	1
7.1.1.2	0.00	0.00	0.00	1
7.1.1.8	1.00	1.00	1.00	1
7.1.1.9	0.00	0.00	0.00	1
7.1.2.2	0.00	0.00	0.00	1
7.2.1.1	0.00	0.00	0.00	1
accuracy			0.44	128
macro avg	0.38	0.44	0.39	128
weighted avg	0.38	0.44	0.39	128

```

/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/s
klearn/linear_model/_sag.py:348: ConvergenceWarning: The max_iter wa
s reached which means the coef_ did not converge
  warnings.warn(
/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/s
klearn/metrics/_classification.py:1565: UndefinedMetricWarning: Prec
ision is ill-defined and being set to 0.0 in labels with no predi
c
ted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(resu
lt))
/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/s
klearn/metrics/_classification.py:1565: UndefinedMetricWarning: Prec
ision is ill-defined and being set to 0.0 in labels with no predi
c
ted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(resu
lt))
/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/s
klearn/metrics/_classification.py:1565: UndefinedMetricWarning: Prec
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c
ted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(resu
lt))

```

```

In [ ]: # Evaluate Predictions
accuracy = accuracy_score(test_labels, y_pred)
f1 = f1_score(test_labels, y_pred, average='weighted') # weighted

print("Accuracy:", accuracy * 100, "%")
print("F1-Score:", f1)

print("\nDetailed classification report:")
print(classification_report(test_labels, y_pred))

# Visualize confusion matrix
ConfusionMatrixDisplay.from_estimator(model, X_test_embed, test_lab
plt.xticks(rotation=90)
plt.show()

# Plot F1-Score und Accuracy
f1 = f1_score(test_labels, y_pred, average='weighted')
acc = accuracy_score(test_labels, y_pred)

```

```
plt.figure(figsize=(6,4))
plt.bar(['F1-Score', 'Accuracy'], [f1, acc], color=['skyblue', 'lightcoral'])
plt.ylim(0,1)
plt.title('Model Performance')
plt.ylabel('Score')
plt.show()

print("F1-Score:", f1)
print("Accuracy:", acc)
```

Accuracy: 43.75 %

F1-Score: 0.39415922619047616

Detailed classification report:

	precision	recall	f1-score	support
1.14.14.18	0.00	0.00	0.00	1
1.15.1.1	1.00	1.00	1.00	1
1.16.3.1	0.00	0.00	0.00	1
1.17.4.1	1.00	1.00	1.00	1
1.5.98.3	0.00	0.00	0.00	1
1.8.3.2	0.00	0.00	0.00	1
2.1.1.354	0.00	0.00	0.00	1
2.1.1.360	0.00	0.00	0.00	1
2.1.1.37	0.50	1.00	0.67	1
2.1.1.72	0.00	0.00	0.00	1
2.1.1.77	1.00	1.00	1.00	1
2.1.1.86	0.50	1.00	0.67	1
2.1.3.15	0.00	0.00	0.00	1
2.3.1.225	0.00	0.00	0.00	1
2.3.1.269	1.00	1.00	1.00	1
2.3.1.48	0.00	0.00	0.00	1
2.3.1.51	1.00	1.00	1.00	1
2.3.2.23	0.00	0.00	0.00	1
2.3.2.26	0.17	1.00	0.29	1
2.3.2.27	0.00	0.00	0.00	1
2.3.2.31	0.00	0.00	0.00	1
2.4.1.109	1.00	1.00	1.00	1
2.4.1.198	0.33	1.00	0.50	1
2.4.2.31	0.00	0.00	0.00	1
2.5.1.18	1.00	1.00	1.00	1
2.7.1.107	0.00	0.00	0.00	1
2.7.1.21	0.00	0.00	0.00	1
2.7.1.33	0.50	1.00	0.67	1
2.7.1.67	0.00	0.00	0.00	1
2.7.10.1	0.50	1.00	0.67	1
2.7.10.2	0.00	0.00	0.00	1
2.7.11.1	0.00	0.00	0.00	1
2.7.11.25	0.00	0.00	0.00	1
2.7.13.3	0.00	0.00	0.00	1
2.7.4.3	1.00	1.00	1.00	1
2.7.6.1	1.00	1.00	1.00	1
2.7.7.108	0.50	1.00	0.67	1
2.7.7.19	0.00	0.00	0.00	1
2.7.7.48	1.00	1.00	1.00	1
2.7.7.49	1.00	1.00	1.00	1

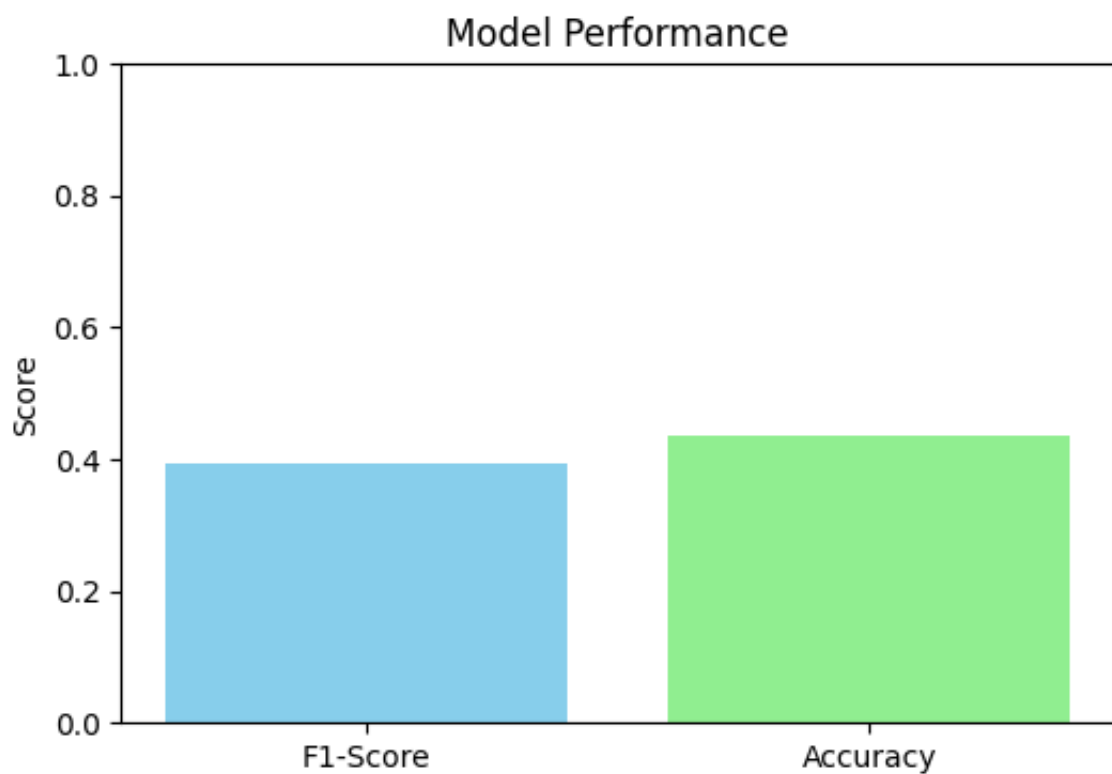
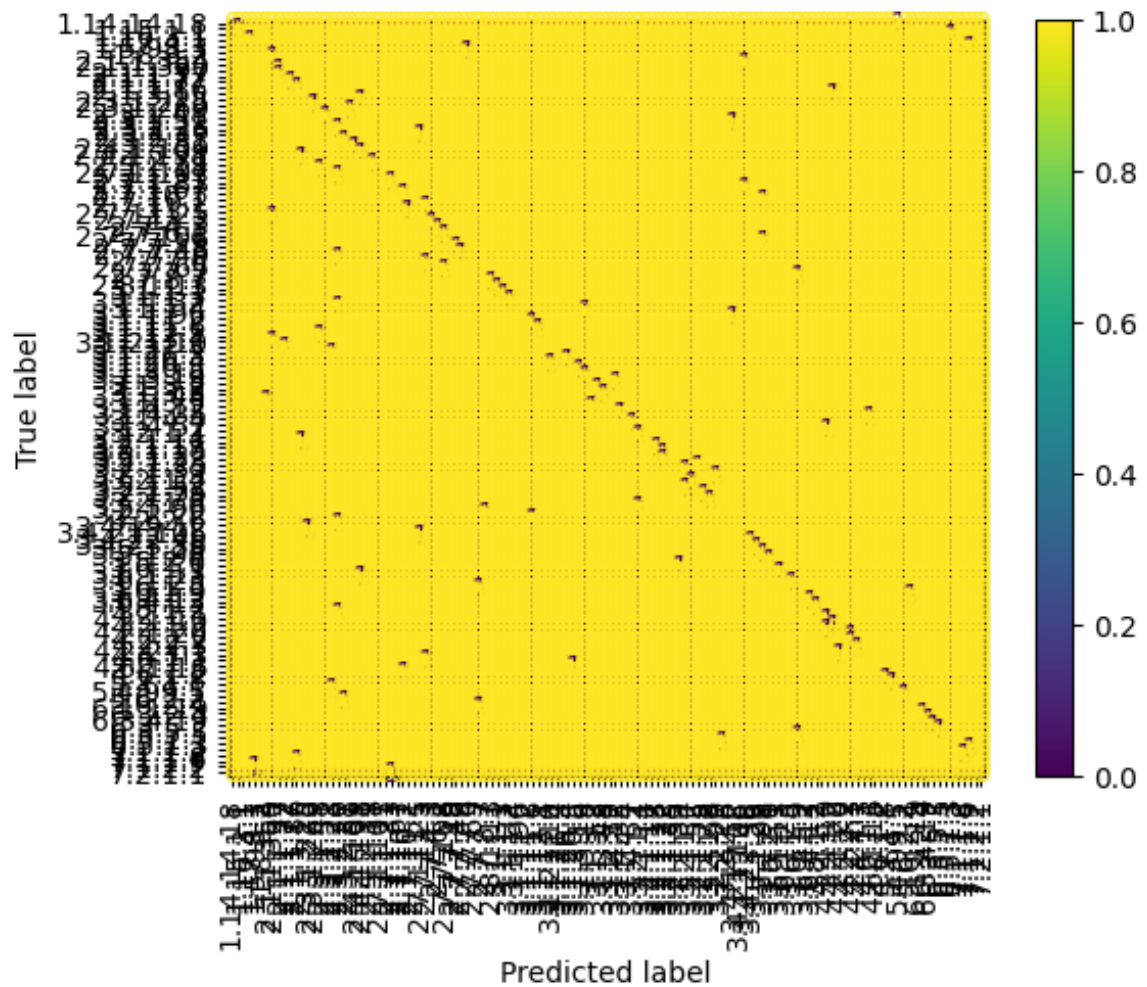
2.7.7.6	0.00	0.00	0.00	1
2.7.7.65	0.00	0.00	0.00	1
2.7.7.7	0.00	0.00	0.00	1
2.7.8.7	0.00	0.00	0.00	1
2.8.1.13	1.00	1.00	1.00	1
3.1.1.1	1.00	1.00	1.00	1
3.1.1.3	1.00	1.00	1.00	1
3.1.1.32	1.00	1.00	1.00	1
3.1.1.4	0.00	0.00	0.00	1
3.1.1.96	0.00	0.00	0.00	1
3.1.11.2	0.00	0.00	0.00	1
3.1.11.6	0.50	1.00	0.67	1
3.1.12.1	1.00	1.00	1.00	1
3.1.13.4	0.00	0.00	0.00	1
3.1.21.10	0.00	0.00	0.00	1
3.1.21.4	0.00	0.00	0.00	1
3.1.26.3	0.00	0.00	0.00	1
3.1.26.4	1.00	1.00	1.00	1
3.1.26.5	0.00	0.00	0.00	1
3.1.3.16	1.00	1.00	1.00	1
3.1.3.18	0.50	1.00	0.67	1
3.1.3.2	0.00	0.00	0.00	1
3.1.3.3	1.00	1.00	1.00	1
3.1.3.48	1.00	1.00	1.00	1
3.1.3.5	0.00	0.00	0.00	1
3.1.4.12	0.00	0.00	0.00	1
3.1.4.35	1.00	1.00	1.00	1
3.1.4.4	0.00	0.00	0.00	1
3.1.4.52	1.00	1.00	1.00	1
3.2.1.1	0.00	0.00	0.00	1
3.2.1.14	0.00	0.00	0.00	1
3.2.1.17	0.00	0.00	0.00	1
3.2.1.18	1.00	1.00	1.00	1
3.2.1.22	0.50	1.00	0.67	1
3.2.1.23	0.00	0.00	0.00	1
3.2.1.35	0.00	0.00	0.00	1
3.2.1.39	0.00	0.00	0.00	1
3.2.1.4	0.00	0.00	0.00	1
3.2.1.52	1.00	1.00	1.00	1
3.2.1.55	0.00	0.00	0.00	1
3.2.1.78	1.00	1.00	1.00	1
3.2.1.8	1.00	1.00	1.00	1
3.2.1.96	0.00	0.00	0.00	1
3.2.2.22	0.00	0.00	0.00	1
3.2.2.6	0.00	0.00	0.00	1
3.4.19.12	0.00	0.00	0.00	1
3.4.21.105	0.00	0.00	0.00	1
3.4.22.49	0.00	0.00	0.00	1
3.4.23.36	1.00	1.00	1.00	1
3.5.1.28	1.00	1.00	1.00	1
3.5.1.98	0.33	1.00	0.50	1
3.5.2.6	1.00	1.00	1.00	1
3.6.1.1	0.00	0.00	0.00	1
3.6.1.13	1.00	1.00	1.00	1
3.6.1.23	0.00	0.00	0.00	1
3.6.1.5	1.00	1.00	1.00	1

3.6.1.9	0.00	0.00	0.00	1
3.6.4.12	0.00	0.00	0.00	1
3.6.4.13	1.00	1.00	1.00	1
3.8.1.2	1.00	1.00	1.00	1
4.2.1.1	0.00	0.00	0.00	1
4.2.1.10	0.33	1.00	0.50	1
4.2.1.2	0.50	1.00	0.67	1
4.2.1.20	0.00	0.00	0.00	1
4.2.2.2	0.00	0.00	0.00	1
4.2.2.3	0.50	1.00	0.67	1
4.2.2.n1	1.00	1.00	1.00	1
4.4.1.3	0.00	0.00	0.00	1
4.6.1.1	0.00	0.00	0.00	1
4.6.1.16	0.00	0.00	0.00	1
4.6.1.2	0.00	0.00	0.00	1
5.2.1.8	1.00	1.00	1.00	1
5.3.4.1	1.00	1.00	1.00	1
5.4.99.5	0.00	0.00	0.00	1
5.6.2.2	1.00	1.00	1.00	1
5.6.2.4	0.00	0.00	0.00	1
6.1.1.19	0.00	0.00	0.00	1
6.3.2.4	1.00	1.00	1.00	1
6.3.4.19	1.00	1.00	1.00	1
6.3.5.2	1.00	1.00	1.00	1
6.3.5.5	1.00	1.00	1.00	1
6.5.1.1	0.00	0.00	0.00	1
6.5.1.3	0.00	0.00	0.00	1
7.1.1.2	0.00	0.00	0.00	1
7.1.1.8	1.00	1.00	1.00	1
7.1.1.9	0.00	0.00	0.00	1
7.1.2.2	0.00	0.00	0.00	1
7.2.1.1	0.00	0.00	0.00	1
accuracy			0.44	128
macro avg	0.38	0.44	0.39	128
weighted avg	0.38	0.44	0.39	128

```

/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/s
klearn/metrics/_classification.py:1565: UndefinedMetricWarning: Prec
ision is ill-defined and being set to 0.0 in labels with no predicte
d samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(resu
lt))
/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/s
klearn/metrics/_classification.py:1565: UndefinedMetricWarning: Prec
ision is ill-defined and being set to 0.0 in labels with no predicte
d samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(resu
lt))
/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/s
klearn/metrics/_classification.py:1565: UndefinedMetricWarning: Prec
ision is ill-defined and being set to 0.0 in labels with no predicte
d samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(resu
lt))

```



F1-Score: 0.39415922619047616

Accuracy: 0.4375

Custom Data Set

```
In [10]: model_customDS = LogisticRegression(max_iter=10, solver='saga', multi_class='multinomial')
model_customDS.fit(X_train_customDS, y_train_customDS)
```

/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/sklearn/linear_model/_logistic.py:1247: FutureWarning: 'multi_class' was deprecated in version 1.5 and will be removed in 1.7. From then on, it will always use 'multinomial'. Leave it to its default value to avoid this warning.

warnings.warn([Parallel(n_jobs=-1)]: Using backend ThreadingBackend with 8 concurrent workers.)

Epoch 1, change: 1
 Epoch 2, change: 0.21424115
 Epoch 3, change: 0.14630157
 Epoch 4, change: 0.11512263
 Epoch 5, change: 0.082461947
 Epoch 6, change: 0.061499406
 Epoch 7, change: 0.062505797
 Epoch 8, change: 0.042685518
 Epoch 9, change: 0.031628279
 max_iter reached after 1089 secondsEpoch 10, change: 0.033246014

/Users/jonathanwolpert/Library/Python/3.9/lib/python/site-packages/sklearn/linear_model/_sag.py:348: ConvergenceWarning: The max_iter was reached which means the coef_ did not converge

warnings.warn(

```
Out[10]: LogisticRegression
LogisticRegression(max_iter=10, multi_class='multinomial',
n_jobs=-1,
solver='saga', verbose=1)
```

```
In [11]: y_pred_customDS = model_customDS.predict(X_test_customDS)
print(classification_report(y_test_customDS, y_pred_customDS))
```

	precision	recall	f1-score	support
1.-.-.-	0.41	0.52	0.46	305
1.1.1.86	1.00	1.00	1.00	127
1.1.1.94	0.98	0.96	0.97	119
1.14.-.-	0.81	0.72	0.76	165
2.1.1.-	0.55	0.68	0.61	465
2.1.1.177	1.00	0.97	0.98	124
2.1.1.182	1.00	0.91	0.95	125
2.1.1.192	0.97	0.87	0.92	127
2.1.1.199	0.99	0.95	0.97	168
2.1.1.228	1.00	0.87	0.93	141
2.1.2.1	1.00	0.97	0.99	159
2.1.2.9	1.00	0.96	0.98	136
2.1.3.15	0.98	0.96	0.97	195
2.1.3.2	1.00	0.94	0.97	141
2.3.1.-	0.49	0.59	0.54	240
2.3.1.234	1.00	0.94	0.97	150
2.3.2.27	0.59	0.61	0.60	270

2.4.1.-	0.45	0.41	0.43	177
2.4.1.227	1.00	0.98	0.99	138
2.4.2.7	0.98	0.97	0.98	129
2.4.2.9	1.00	0.96	0.98	140
2.5.1.19	0.98	0.94	0.96	118
2.5.1.6	0.99	0.99	0.99	156
2.5.1.75	0.97	0.93	0.95	158
2.5.1.78	0.98	0.96	0.97	123
2.7.1.148	0.98	0.94	0.96	117
2.7.1.33	0.98	0.92	0.95	127
2.7.11.1	0.48	0.63	0.54	484
2.7.2.3	1.00	0.96	0.98	140
2.7.4.3	0.99	0.93	0.96	174
2.7.4.6	1.00	0.98	0.99	143
2.7.4.9	0.98	0.97	0.98	123
2.7.7.3	1.00	0.98	0.99	135
2.7.7.6	0.87	0.91	0.89	789
2.7.7.7	0.78	0.70	0.74	140
2.7.7.8	1.00	0.96	0.98	157
2.7.8.13	1.00	0.96	0.98	143
2.7.8.7	1.00	0.92	0.96	119
2.8.1.13	1.00	0.94	0.97	145
2.8.1.6	0.99	0.85	0.92	117
2.8.1.8	0.93	0.81	0.87	129
2.8.4.3	0.99	0.90	0.94	123
3.1.-.-	0.63	0.80	0.71	780
3.1.1.29	0.99	0.96	0.97	156
3.1.1.96	1.00	0.93	0.96	120
3.1.11.6	0.98	0.95	0.97	171
3.1.21.10	0.99	0.95	0.97	137
3.1.26.4	0.93	0.87	0.90	228
3.1.26.5	0.94	0.87	0.90	141
3.1.3.16	0.70	0.58	0.64	117
3.4.21.-	0.69	0.64	0.66	198
3.4.21.92	0.97	0.98	0.98	151
3.4.24.-	0.59	0.66	0.62	278
3.5.1.5	0.99	1.00	1.00	164
3.6.-.-	0.97	0.90	0.93	153
3.6.1.-	0.90	0.84	0.87	215
3.6.1.27	0.99	0.98	0.98	128
3.6.4.-	0.80	0.76	0.78	343
3.6.4.13	0.63	0.68	0.65	290
3.6.5.-	0.93	0.82	0.87	293
3.6.5.n1	1.00	1.00	1.00	141
4.2.1.11	1.00	0.99	1.00	182
4.2.1.20	0.99	0.92	0.96	199
4.2.1.33	1.00	0.99	0.99	211
4.2.3.5	1.00	0.99	0.99	142
4.3.2.1	1.00	0.93	0.97	123
4.3.2.10	1.00	0.99	1.00	124
5.2.1.8	0.88	0.81	0.85	296
5.3.1.1	1.00	0.94	0.97	123
5.3.1.16	0.99	0.98	0.99	124
5.3.1.9	1.00	0.91	0.95	126
5.4.2.10	0.99	0.97	0.98	137
5.4.3.8	1.00	0.97	0.98	135

5.4.99.12	1.00	0.91	0.95	125
5.6.1.7	1.00	1.00	1.00	202
6.1.1.11	1.00	0.96	0.98	164
6.1.1.14	1.00	0.96	0.98	169
6.1.1.15	1.00	0.96	0.98	167
6.1.1.16	0.99	0.96	0.98	148
6.1.1.17	1.00	0.90	0.95	175
6.1.1.19	1.00	0.93	0.96	149
6.1.1.20	0.98	0.94	0.96	202
6.1.1.21	1.00	0.98	0.99	141
6.1.1.3	1.00	0.94	0.97	140
6.1.1.4	1.00	0.89	0.94	153
6.1.1.7	1.00	0.95	0.97	141
6.3.2.8	0.99	0.99	0.99	135
6.3.4.2	1.00	0.99	0.99	138
6.3.4.4	1.00	0.94	0.97	165
6.3.4.5	1.00	1.00	1.00	121
6.3.5.-	0.94	0.89	0.91	251
6.3.5.2	1.00	0.92	0.96	132
6.3.5.7	0.99	0.94	0.97	122
6.5.1.2	0.98	0.93	0.95	175
7.-.-.-	0.97	0.90	0.93	164
7.1.1.-	0.90	0.93	0.92	925
7.1.1.2	0.99	0.91	0.95	258
7.1.1.9	1.00	0.80	0.89	139
7.1.2.2	0.99	0.98	0.99	422
7.4.2.8	1.00	0.93	0.97	168
accuracy			0.88	19118
macro avg	0.93	0.90	0.91	19118
weighted avg	0.89	0.88	0.88	19118

Safe model

```
In [12]: np.save("X_train_embed.npy", X_train_customDS)
np.save("X_test_embed.npy", X_test_customDS)
```

Load model

```
In [ ]: X_train_embed = np.load("X_train_embed.npy")
X_test_embed = np.load("X_test_embed.npy")
```

Evaluate Predictions

```
In [ ]: # Evaluate Predictions
accuracy = accuracy_score(y_test_customDS, y_pred_customDS)
f1 = f1_score(y_test_customDS, y_pred_customDS, average='weighted')

print("Accuracy:", accuracy * 100, "%")
print("F1-Score:", f1)

print("\nDetailed classification report:")
```

```
print(classification_report(y_test_customDS, y_pred_customDS))

# Visualize confusion matrix
ConfusionMatrixDisplay.from_estimator(model_customDS, X_test_customDS,
plt.xticks(rotation=90))
plt.show()
```

Accuracy: 87.54576838581441 %

F1-Score: 0.8807537462237521

Detailed classification report:

	precision	recall	f1-score	support
1.-.-.-	0.41	0.52	0.46	305
1.1.1.86	1.00	1.00	1.00	127
1.1.1.94	0.98	0.96	0.97	119
1.14.-.-	0.81	0.72	0.76	165
2.1.1.-	0.55	0.68	0.61	465
2.1.1.177	1.00	0.97	0.98	124
2.1.1.182	1.00	0.91	0.95	125
2.1.1.192	0.97	0.87	0.92	127
2.1.1.199	0.99	0.95	0.97	168
2.1.1.228	1.00	0.87	0.93	141
2.1.2.1	1.00	0.97	0.99	159
2.1.2.9	1.00	0.96	0.98	136
2.1.3.15	0.98	0.96	0.97	195
2.1.3.2	1.00	0.94	0.97	141
2.3.1.-	0.49	0.59	0.54	240
2.3.1.234	1.00	0.94	0.97	150
2.3.2.27	0.59	0.61	0.60	270
2.4.1.-	0.45	0.41	0.43	177
2.4.1.227	1.00	0.98	0.99	138
2.4.2.7	0.98	0.97	0.98	129
2.4.2.9	1.00	0.96	0.98	140
2.5.1.19	0.98	0.94	0.96	118
2.5.1.6	0.99	0.99	0.99	156
2.5.1.75	0.97	0.93	0.95	158
2.5.1.78	0.98	0.96	0.97	123
2.7.1.148	0.98	0.94	0.96	117
2.7.1.33	0.98	0.92	0.95	127
2.7.11.1	0.48	0.63	0.54	484
2.7.2.3	1.00	0.96	0.98	140
2.7.4.3	0.99	0.93	0.96	174
2.7.4.6	1.00	0.98	0.99	143
2.7.4.9	0.98	0.97	0.98	123
2.7.7.3	1.00	0.98	0.99	135
2.7.7.6	0.87	0.91	0.89	789
2.7.7.7	0.78	0.70	0.74	140
2.7.7.8	1.00	0.96	0.98	157
2.7.8.13	1.00	0.96	0.98	143
2.7.8.7	1.00	0.92	0.96	119
2.8.1.13	1.00	0.94	0.97	145
2.8.1.6	0.99	0.85	0.92	117
2.8.1.8	0.93	0.81	0.87	129
2.8.4.3	0.99	0.90	0.94	123
3.1.-.-	0.63	0.80	0.71	780

3.1.1.29	0.99	0.96	0.97	156
3.1.1.96	1.00	0.93	0.96	120
3.1.11.6	0.98	0.95	0.97	171
3.1.21.10	0.99	0.95	0.97	137
3.1.26.4	0.93	0.87	0.90	228
3.1.26.5	0.94	0.87	0.90	141
3.1.3.16	0.70	0.58	0.64	117
3.4.21.-	0.69	0.64	0.66	198
3.4.21.92	0.97	0.98	0.98	151
3.4.24.-	0.59	0.66	0.62	278
3.5.1.5	0.99	1.00	1.00	164
3.6.-.-	0.97	0.90	0.93	153
3.6.1.-	0.90	0.84	0.87	215
3.6.1.27	0.99	0.98	0.98	128
3.6.4.-	0.80	0.76	0.78	343
3.6.4.13	0.63	0.68	0.65	290
3.6.5.-	0.93	0.82	0.87	293
3.6.5.n1	1.00	1.00	1.00	141
4.2.1.11	1.00	0.99	1.00	182
4.2.1.20	0.99	0.92	0.96	199
4.2.1.33	1.00	0.99	0.99	211
4.2.3.5	1.00	0.99	0.99	142
4.3.2.1	1.00	0.93	0.97	123
4.3.2.10	1.00	0.99	1.00	124
5.2.1.8	0.88	0.81	0.85	296
5.3.1.1	1.00	0.94	0.97	123
5.3.1.16	0.99	0.98	0.99	124
5.3.1.9	1.00	0.91	0.95	126
5.4.2.10	0.99	0.97	0.98	137
5.4.3.8	1.00	0.97	0.98	135
5.4.99.12	1.00	0.91	0.95	125
5.6.1.7	1.00	1.00	1.00	202
6.1.1.11	1.00	0.96	0.98	164
6.1.1.14	1.00	0.96	0.98	169
6.1.1.15	1.00	0.96	0.98	167
6.1.1.16	0.99	0.96	0.98	148
6.1.1.17	1.00	0.90	0.95	175
6.1.1.19	1.00	0.93	0.96	149
6.1.1.20	0.98	0.94	0.96	202
6.1.1.21	1.00	0.98	0.99	141
6.1.1.3	1.00	0.94	0.97	140
6.1.1.4	1.00	0.89	0.94	153
6.1.1.7	1.00	0.95	0.97	141
6.3.2.8	0.99	0.99	0.99	135
6.3.4.2	1.00	0.99	0.99	138
6.3.4.4	1.00	0.94	0.97	165
6.3.4.5	1.00	1.00	1.00	121
6.3.5.-	0.94	0.89	0.91	251
6.3.5.2	1.00	0.92	0.96	132
6.3.5.7	0.99	0.94	0.97	122
6.5.1.2	0.98	0.93	0.95	175
7.-.-.-	0.97	0.90	0.93	164
7.1.1.-	0.90	0.93	0.92	925
7.1.1.2	0.99	0.91	0.95	258
7.1.1.9	1.00	0.80	0.89	139
7.1.2.2	0.99	0.98	0.99	422

7.4.2.8	1.00	0.93	0.97	168
accuracy			0.88	19118
macro avg	0.93	0.90	0.91	19118
weighted avg	0.89	0.88	0.88	19118

